UPM PUTRA PERTANIAN UNTUK RAKYAT FAKULTI KEJURUTERAAN

WasteQuest

The Waste-to-Wealth Book

Learn it · Play it · Make it



eBook · 2026 edition

EN · BM

WasteQuest

The Waste-to-Wealth Book

Written by

Wan Azlina Wan Ab Karim Ghani, Shafreeza Sobri, Izzudin Ibrahim, Nur Syakina Jamali, Mohamad Faiz Mukhtar Gunam Resul, Salmiaton Ali

Department of Chemical and Environmental Engineering, Faculty of Engineering, Universiti Putra Malaysia

The WasteQuest website, games and videos: **wastequest.github.io**

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📖 How to use this book



Hi! I'm Kitar, the recycling robot of Eco-City. I'll be your guide. Let's turn waste into wealth!

📖 **Turn pages:** swipe, tap the arrows, or use the ← → keys. Tap **Contents** to jump.

▶ **Videos** play right on the page: tap the picture.

✅ **Quick quizzes:** tap an answer to check it.

🔗 **Green buttons** open the matching game or lab on the WasteQuest website.

🧒 **Reading level:** choose Kids, Teens, Adults or Teacher in the toolbar. Lessons, science notes and quizzes change to match.

⬇️ **PDF:** download the whole book to print or share.

Look out for these signs

⛔ Danger: an adult does this step	⚠️ Take care
🔬 The science behind a step	👩‍🏫 Tip for teachers

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Did you know?

Mercury is toxic. A broken lamp releases mercury vapour, so lamps go to hazardous-waste collection.

Part 1

Welcome to Eco-City

A little island town that needs your help.



The story of Eco-City

Eco-City is a little island town. Its people throw away a lot: food scraps, bottles, bags, old clothes and broken gadgets. The rubbish piles up, the sky turns grey and the river starts to smell.



Before: a messy, smoggy town



After: a clean, green town

Every time you learn something, play a game, finish a mission or make something useful from waste, one part of the town gets cleaner. Clean all six districts and Eco-City shines again!

🗺️ The six districts

Each district of Eco-City is one part of WasteQuest, and one part of this book.



🏫 Academy

Lessons, quick quizzes and your certificate.

Part 2: The Academy: Learn



♻️ Recycling Plant

Sorting and word games, and the camera scan-and-sort helper.

Part 3: Play: Games, Missions & the Arena



🌱 Compost Farm

Compost, eco-enzyme and growing labs.

Part 4: Make: Maker Lab & Compost Farm



🛠️ Maker Lab

Craft and science labs: candles, soap, bags, bricks and more.

Part 4: Make: Maker Lab & Compost Farm



🏪 Market

Money from waste, your footprint and the town shop.

Part 3: Play: Games, Missions & the Arena



🏟️ Arena

Live class battles and Event Town.

Part 3: Play: Games, Missions & the Arena

👋 Who is this book for?

Everyone! Pick your reading level in the toolbar and the book changes to match.



Kids 7-12

Start with the story, play the games in Part 3 and try the no-heat labs in Part 4 (look for “No heat”). Do every hot or sharp step with an adult.



Teens 13-17

Read the lessons in Part 2 with the science notes, take the quick quizzes and aim for the Zero-Waste Champion certificate.



Adults

Community groups, university students and company staff: the labs, the business and ESG notes and the Waste-to-Wealth Practitioner certificate are for you.



Teacher

Teachers and facilitators: Part 6 has session plans, safety rules, rubrics, curriculum links and how to run Class Battle. Teacher tips appear in every lab.

💎 What is waste-to-wealth?

Waste-to-wealth (W2W) means seeing rubbish as a resource. Used cooking oil can become a candle or soap, fruit peels become eco-enzyme, food scraps become compost and plastic becomes bricks, bags and planters.

🌍 Environment	💰 Economy	🤝 Society
Less rubbish in landfills, drains, rivers and the sea.	Free materials become products people value, and new small businesses.	Skills, teamwork and cleaner, healthier neighbourhoods.

WasteQuest supports the UN Sustainable Development Goals, especially SDG 12 (responsible consumption and production), SDG 13 (climate action) and SDG 15 (life on land).





Did you know?

Fresh peels ferment well. Meat and fish would rot and smell bad.

Part 2

The Academy: Learn

Seven short lessons on where waste goes and how to turn it into something useful.





Lesson 1



Where does our waste go?

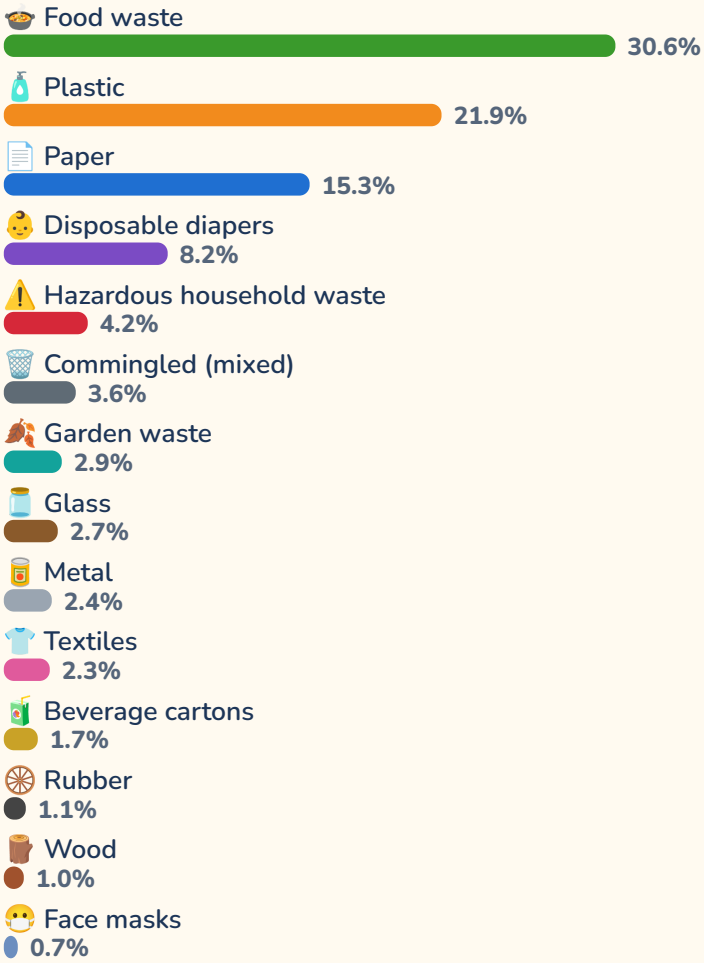
Malaysia generates about 39,078 tonnes of solid waste every day (SWCorp, 2024). Most of it still ends up in landfills, where it takes space, pollutes and releases greenhouse gases.

39,078 t
of solid waste per day in Malaysia (2024)

1.17 kg
per person, per day

×2
daily waste roughly doubled from 2005 to 2024

What is in a Malaysian household bin?



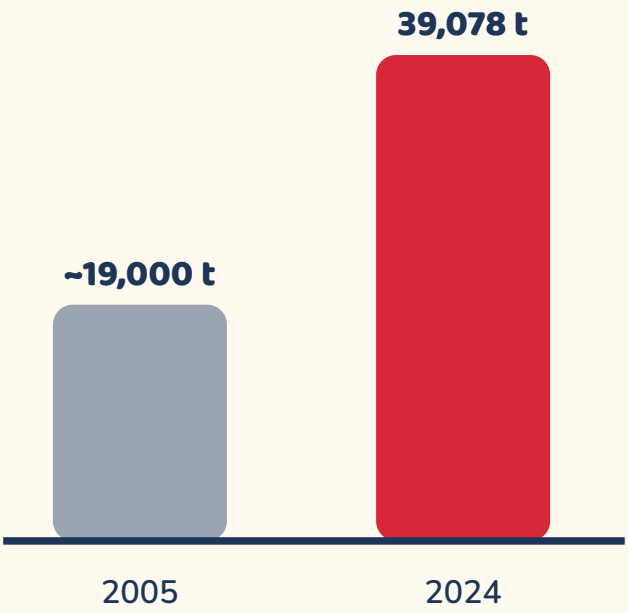
Household waste composition, % by weight. Source: SWCorp via The Star, 2 Jan 2024.

Your 1.17 kg a day, stacked up



Rough split using the household mix above.

Daily waste: 2005 vs 2024



💡 Food (30.6%), plastic (21.9%) and paper (15.3%) make up about two-thirds of household waste, and all three can be composted or recycled.

🔬 The science

In a landfill, buried food waste has no oxygen. Anaerobic microbes break it down into **landfill gas**, roughly 50% methane (CH_4) and 50% carbon dioxide (US EPA). For sugar, a simplified equation is **$C_6H_{12}O_6 \rightarrow 3\ CH_4 + 3\ CO_2$**.

Methane traps far more heat than CO_2: its global warming potential over 100 years is about **28** (IPCC AR5, 2013). So 1 kg of methane ≈ 28 kg CO_2-equivalent. Composting with air (aerobic) gives mainly CO_2 and water instead.

Maths: 1.17 kg × 365 days ≈ **427 kg** per person per year.

💼 ESG and business angle

Malaysia's national recycling rate was reported as 37.9% in 2024 (KPKT, via Bernama 2025); experts note this counts recyclables collected, not necessarily recycled. For organisations, waste is a standard ESG disclosure topic (e.g. GRI 306: Waste 2020): tonnes generated, diverted from disposal and sent to disposal.

🧑‍🏫 **Teacher tip:** 15 min. Ask: “What did you throw away today?” before showing the chart. Let groups guess the top 3 materials, then reveal. Follow with Sort It Out (#/game/sort).

✅ Quick quiz

1. Why is food waste in a landfill a climate problem?

A. It decomposes without oxygen and releases methane

B. It turns into plastic

C. It absorbs carbon dioxide

2. Daily solid waste in Malaysia went from ~19,000 t (2005) to ~39,000 t (2024). That is an increase of about…

A. 20%

B. 50%

C. 100% (it doubled)

3. What is the biggest part of Malaysian household waste?

A. Plastic

B. Food waste

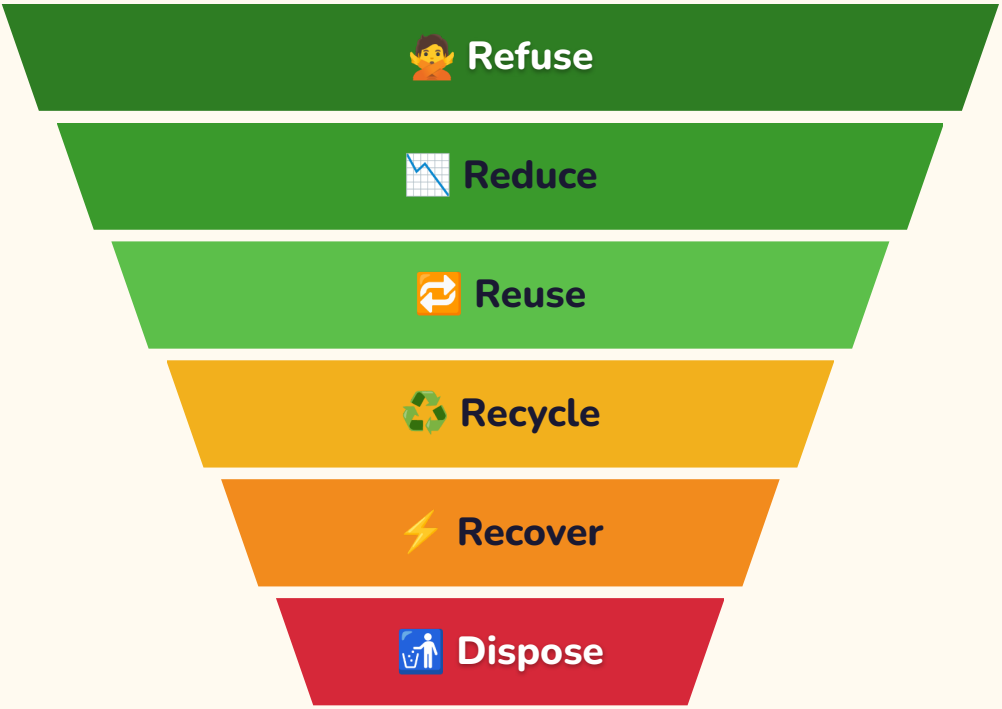
C. Glass

Answers: 1 A · 2 C · 3 B

Lesson 2

🔺 The waste hierarchy

The waste hierarchy ranks options from most to least preferred. The higher the step, the more value we keep and the less energy and raw material we use.



1. 🙅 **Refuse:** Stop waste before it exists: turn down single-use items and freebies you will not use. *Plastic straws, extra plastic bags, flyers, free pens you will never use.*
2. 📉 **Reduce:** Use fewer resources: buy less, choose less packaging and plan meals so food is not wasted. *Refill bottles, print double-sided, buy in bulk, take smaller food portions.*

3. 🔁 **Reuse:** Keep products in use for longer: repair, share, donate or repurpose them (upcycling). *Old T-shirt to tote bag, bottle to planter, jam jar to pencil holder.*
4. ♻️ **Recycle:** Turn waste materials back into raw materials: paper, plastic, metal and glass go to the SWCorp bins; food waste can be composted or fermented. *Paper to new paper, cans to new cans, peels to compost or eco-enzyme, used cooking oil to candles.*
5. ⚡ **Recover:** Recover energy from what is left, for example biogas from food waste or electricity from waste-to-energy plants. *Biogas digesters, waste-to-energy plants.*
6. 🚮 **Dispose:** Safe disposal in a sanitary landfill is the last resort: all the value in the material is lost. *Used tissues, dirty masks, broken mirrors (wrapped safely).*

🔬 **The science**

Why is higher better? Each step down loses more of the energy and materials already invested. Example: recycling aluminium cans saves more than 90% of the energy needed to make new aluminium from bauxite ore (The Aluminum Association), but refusing the can saves 100%.

“Refuse” comes from the zero-waste movement (Bea Johnson, 2013); official hierarchies such as the EU Waste Framework Directive start at prevention.

💼 ESG and business angle

Business translation: Refuse/Reduce = redesign packaging and processes; Reuse = refill and take-back schemes, repair services; Recycle = closed-loop material contracts; Recover = biogas or energy from residues. Moving up the hierarchy usually cuts disposal fees as well as emissions.

🧑‍🏫 **Teacher tip:** 10 min. Give each group a waste item card and ask them to place it on the pyramid. Discuss: why is recycling not the top?

✅ Quick quiz

1. Put in order, best first: Recycle, Reduce, Dispose

A. Recycle → Reduce → Dispose

B. Reduce → Recycle → Dispose

C. Dispose → Recycle → Reduce

2. Turning an old T-shirt into a tote bag is an example of...

A. Reuse (upcycling)

B. Recover

C. Dispose

3. Which is the BEST choice for a plastic straw you don't need?

A. Recycle it after use

B. Say no to it (refuse)

C. Throw it away

Answers: 1 B · 2 A · 3 B

Lesson 3

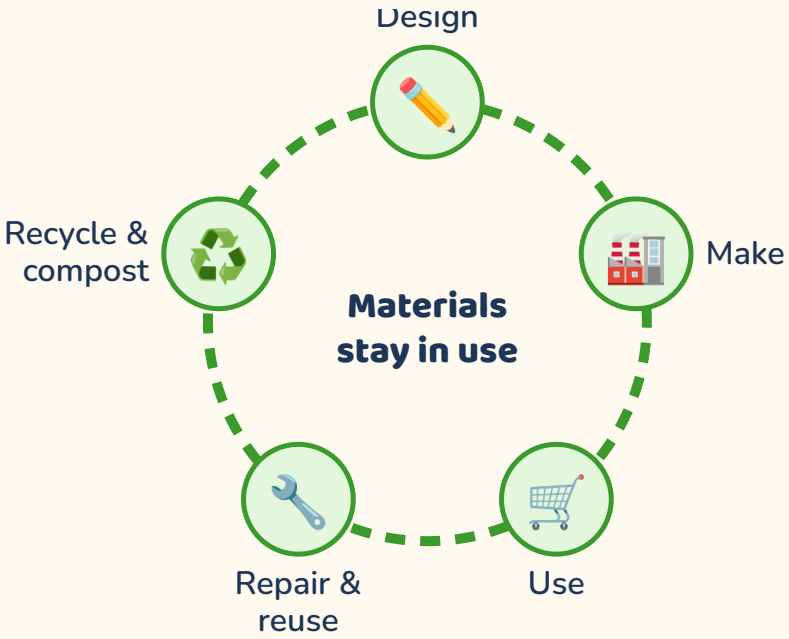
🔄 Linear vs circular economy

A linear economy follows “take, make, use, throw away”. A circular economy designs waste out, keeps products and materials in use at their highest value, and regenerates nature (Ellen MacArthur Foundation).

Linear



Circular



	➡️ Linear	🔄 Circular
Design	Cheap, single-use, hard to repair	Durable, repairable, easy to take apart and recycle
End of life	Landfill or open burning	Becomes a resource for a new product, or compost
Example	Used cooking oil poured down the sink	Used cooking oil made into candles (**Eco-Candles from Used Cooking Oil**)

🔬 **The science**

Circular thinking uses two loops. The **biological cycle**: food and plant materials go back to the soil safely (composting, eco-enzyme). The **technical cycle**: metals, plastics and glass are kept in use by reuse, repair, remanufacture and recycling. Problems start when the two are mixed, such as food stuck to plastic.

💼 **ESG and business angle**

Circular business models: product-as-a-service (leasing), refill and reuse systems, take-back and remanufacture, and selling by-products. **Extended Producer Responsibility (EPR)** makes producers responsible for their packaging after use. In Malaysia, KPKT has announced a voluntary EPR phase for packaging from 2026, becoming mandatory by 2030 (Bernama, 2025).

👩‍🏫 **Teacher tip:** 10 min. Toggle the diagram on the projector. Ask learners to trace one product (a drink bottle) through both systems.

✅ Quick quiz

1. Which is one of the three circular economy principles?

A. Make products cheaper to throw away

B. Keep products and materials in use

C. Burn all waste for energy

2. Compost belongs to which loop?

A. Biological cycle

B. Technical cycle

3. In a circular economy, what happens to old things?

A. They become useful again

B. They go to the landfill

Answers: 1 B · 2 A · 3 A

Lesson 4

💎 Waste-to-Wealth (W2W)

Waste-to-Wealth (W2W) turns waste into products, materials or energy with value. It is the circular economy in action: it reduces, reuses, recycles and recovers, and extends the life of products.

Three kinds of benefit

🌍 Environmental

- Less pollution of air, rivers and sea
- Lower greenhouse gas emissions (less methane from landfills)
- Conserves raw materials, water and energy

💰 Economic

- Value-added products made from cheap or free materials
- Green entrepreneurship and small businesses
- New jobs and income opportunities

🤝 Social

- Greater awareness in schools and communities
- Sustainable lifestyles and habits
- Innovation, creativity and teamwork

From waste to wealth: our 18 labs

Used cooking oil
🕯️ **Eco-Candles from Used Cooking Oil** →

Fish waste
🐟 **Fish-Waste Pet Treats** →

Used oil, old T-shirts and bottle caps
🎁 **Trash to Treasure: Soap, Tote & Toy Car** →

Coloured plant scraps
🌈 **Say No to Litmus: Rainbow pH from Kitchen Waste** →

Used coffee grounds
☕ **Kopi Wira Bau: Coffee-Ground Odour Absorbers** →

Plastic bottles
🌱 **Self-Watering Bottle Garden** →

Fruit and vegetable peels
🍊 **Eco-Enzyme** →

Soft plastic wrappers
🧱 **Eco-Bricks & Eco-Bags** →

Food and garden waste
🪱 **Composting: Bokashi & Worms** →

Plant starch (instead of plastic)
🌽 **Cornstarch Bioplastic** →

Plastic bottles and compost
🪴 **Bottle Vertical Garden** →

Plastic bottles
🥬 **Bottle Hydroponics** →

Plastic bags and straws
☂️ **Fused-Plastic Bags & Mini Umbrella** →

Bubble wrap
🛟 **Bubble-Wrap Float Ring (Model)** →

Bubble wrap and old cloth
🛌 **3-in-1 Bubble-Wrap Sleeping Bag** →

🔬 **The science**

Behind each product is real science: microbes turn food waste into compost (compost sim); fermentation makes eco-enzyme acidic (eco-enzyme sim); plant pigments called anthocyanins change colour with pH (pH lab); glycerol acts as a plasticiser to make starch bioplastic flexible.

💼 **ESG and business angle**

Green entrepreneurship starts with cheap inputs. UPM ENG3104 student teams (2024) estimated a starch bioplastic pot at about RM0.84 in materials and a tote bag from used plastic bags at about RM4.66 (student estimates, not market prices). Test your own pricing in Waste to Cash. A credible W2W business also needs a safe process, a steady waste supply, quality control and honest environmental claims.

👩‍🏫 **Teacher tip:** 15 min. Flip the three cards with the class, then let each group choose the lab they will do in the hands-on session.

✅ Quick quiz

1. Which is a SOCIAL benefit of W2W?

A. Lower greenhouse gas emissions

B. Greater community awareness and new skills

C. Higher product prices

2. Fruit peels + sugar + water, fermented, make...

A. Eco-enzyme

B. Bioplastic

C. Candles

3. What does Waste-to-Wealth mean?

A. Turning rubbish into useful things

B. Burying more rubbish

C. Buying new things every day

Answers: 1 B · 2 A · 3 A

Lesson 8

🏛️ Malaysia's rules and bins

Under the Solid Waste and Public Cleansing Management Act 2007 (Act 672), separation at source is mandatory in the states and territories that adopted the Act. Recyclables are separated from other household waste for collection.

Where separation at source is mandatory (Act 672):

Johor · Melaka · Negeri Sembilan · Pahang · Perlis · Kedah · Kuala Lumpur · Putrajaya

Other states run their own programmes. Selangor is working towards adopting the Act in phases; the start date is not yet confirmed, so check your council's rules.

📄 **Blue bin: paper**

✅ Newspapers, magazines, cardboard boxes (flattened), exercise books, envelopes, paper bags.

❌ Used tissues, oily pizza boxes, wet or food-soiled paper.

🥫 Orange bin: plastic & aluminium/metal

✅ Rinsed plastic bottles and containers, aluminium drink cans, food tins, clean plastic bags.

❌ Oily food boxes, used face masks, styrofoam with food left in it.

🫙 Brown bin: glass

✅ Glass bottles and jars (rinsed, lids removed).

❌ Mirrors, window glass, light bulbs, ceramic plates. Wrap broken glass safely.

3 easy steps

1️⃣ Rinse · 2️⃣ Dry · 3️⃣ Separate into the right bin. Food waste and dirty items go with general waste, or better, into compost. Batteries, e-waste and medicine go to special collection points.

🗑️ Play: Sort It Out!

🔬 **The science**

Why rinse? Food and oil left on packaging contaminate other recyclables: wet, greasy paper fibres cannot be pulped well, and dirty plastic lowers the quality of recycled pellets. Clean, dry, separated materials are worth more to recyclers.

💼 **ESG and business angle**

Policy context for organisations: Act 672 (separation at source), Malaysia's Roadmap Towards Zero Single-Use Plastics 2018–2030, and the planned EPR scheme for packaging (voluntary from 2026, mandatory by 2030). Workplaces can mirror the household system with labelled blue, orange and brown bins plus a food-waste stream.

👩‍🏫 **Teacher tip:** 10 min. Bring three coloured boxes and 10 real (clean) items. Then play Sort It Out on the projector, Level 1 then Level 2.

✅ Quick quiz

1. Which law makes separation at source mandatory in adopting states?

A. Act 672 (2007)

B. Environmental Quality Act 1974

C. There is no such law

2. An aluminium drink can goes into the…

A. Blue bin

B. Orange bin

C. Brown bin

3. Which bin is for paper?

A. Blue

B. Orange

C. Brown

Answers: 1 A · 2 B · 3 A

Lesson 6

🎯 The Sustainable Development Goals

In 2015, all UN member states, including Malaysia, adopted 17 Sustainable Development Goals (SDGs) for 2030. Waste-to-Wealth links most directly to SDGs 12, 13 and 15, and also to 11 and 14.



12 Responsible consumption and production

Target 12.5: substantially reduce waste generation through prevention, reduction, recycling and reuse by 2030. Target 12.3: halve per-capita food waste.

- 🪱 **Composting: Bokashi & Worms**
- 🍊 **Eco-Enzyme**
- 🎁 **Trash to Treasure: Soap, Tote & Toy Car**
- 🕯️ **Eco-Candles from Used Cooking Oil**
- 🐟 **Fish-Waste Pet Treats**
- ☕ **Kopi Wira Bau: Coffee-Ground Odour Absorbers**

13

Climate action

Keeping food waste out of landfills cuts methane, a strong greenhouse gas. Recycling saves the energy needed to make new materials.

- 🪱 Composting: Bokashi & Worms
- 🍊 Eco-Enzyme
- 🌽 Cornstarch Bioplastic

15

Life on land

Compost returns nutrients to soil; less landfill means less land cleared and less leachate polluting soil and groundwater.

- 🪱 Composting: Bokashi & Worms
- 🌱 Self-Watering Bottle Garden
- 🌈 Say No to Litmus: Rainbow pH from Kitchen Waste

11

Sustainable cities and communities

Target 11.6: reduce the environmental impact of cities, including better municipal waste management.

- 🧱 Eco-Bricks & Eco-Bags

14

Life below water

Target 14.1: reduce marine pollution, much of which comes from land, such as plastic litter.

🧱 **Eco-Bricks & Eco-Bags**

🎁 **Trash to Treasure: Soap, Tote & Toy Car**

💼 **ESG and business angle**

Organisations often map their ESG reports to the SDGs. A W2W project can report outputs (kg of waste diverted, products made) and link them to Target 12.5 (waste reduction) and Target 12.3 (food waste).

👩‍🏫 **Teacher tip:** 5–10 min. Ask each group to name which SDG their lab supports and why, then add it to their pitch poster.

✅ Quick quiz

1. Composting school food waste instead of landfilling it mainly supports...

A. SDG 12 and SDG 13

B. SDG 1 only

C. SDG 9 only

2. SDG Target 12.3 aims to…

A. Halve per-capita food waste

B. Double plastic production

C. Build more landfills

3. Which SDG is about using things responsibly and wasting less?

A. SDG 12

B. SDG 3

C. SDG 7

Answers: 1 A · 2 A · 3 A

Lesson 7

🏆 Be a Zero-Waste Champion

Build a personal (or team) action plan. Pick specific, measurable actions, set a start date and review after 30 days.

📝 My Zero-Waste action plan

Name

Refuse

- [ ] Say no to plastic straws and extra bags
- [ ] Bring my own water bottle and food container

Reduce

- [ ] Take only the food I can finish
- [ ] Print on both sides of the paper

Reuse

- [ ] Use a reusable shopping bag
- [ ] Give old clothes, toys and books a second life

Recycle

- [ ] Rinse and sort recyclables into the blue, orange and brown bins
- [ ] Start a compost bin or an eco-enzyme bottle

Share

- [ ] Teach my family how to sort waste
- [ ] Join or start an Eco-Club

My own idea

I will start on

What does a Zero-Waste Champion do?

Practises the waste hierarchy daily, leads by example in school, measures progress (e.g. weighs the class bin each week) and shares ideas in the community.

🎮 Games
🧪
🎓 Certificate

👩‍🏫 **Teacher tip:** 10 min. Learners complete and print the plan as an exit ticket. Review the plans after 2–4 weeks (Eco-Club week 4).

✅ Quick quiz

1. Which goal is the most useful for an action plan?

A. “Be greener”

B. “Bring my own bottle every school day for 4 weeks”

2. How can your class measure its progress?

A. Weigh the general-waste bin every week

B. Guess

3. What is the best way to start being a Zero-Waste Champion?

A. Do everything at once, then stop

B. Start with one small habit and keep it

Answers: 1 B · 2 A · 3 B



Did you know?

Raw fish spoils quickly. Proper cooking and cooling make it safer, and bones must be checked.

Part 3

Play: Games, Missions & the Arena

Games, virtual labs, real-world missions, class battles and certificates.



🎮 Games & virtual labs

Every game and simulation earns a badge and coins for your town. Tap **Play** to open it.

🎮 Games

🗑️ **Sort It Out!**
Drag each item into the right Malaysian recycling bin.
Ages 7+

🃏 **Eco-Match**
Flip the cards. Match each waste with the useful product it becomes.
Ages 7+

✏️ **Let's Cross Together!**
Solve a waste-to-wealth crossword. A new puzzle every time.
Ages 8+

🔎 **Fun Puzzle: Word Search**
Find the hidden waste-to-wealth words in the grid.
Ages 7+

🕵️ **Myth or Fact?**
Swipe each card: is it a myth or a fact? Learn the real story behind each one.
Ages 7+

⚡ **Eco Quiz Blast**
Beat the 20-second clock! Fast answers score more points.
Ages 7+

🧪 Simulations: virtual labs

Say No to Litmus!

🧪 Test kitchen liquids with natural indicators from cabbage, bunga telang and kunyit.

Ages 7+

Compost Master

🌱 Build a compost bin, balance greens and browns, and simulate 8 weeks of hot composting.

Ages 7+

Eco-Enzyme Mixer

🍊 Solve the 1:3:10 recipe for your bottle, then fast-forward 3 months of fermentation.

Ages 7+

Trash-to-Cash Tycoon

💰 Run a 10-day waste-to-wealth business: turn neighbourhood waste into products and profit.

Ages 7+

My Waste Footprint

👣 Count a week of your family's waste, compare with Malaysia, then make a pledge card.

Ages 7+

✅ Quick quiz

1. Which SWCorp recycling bin is for paper?

A. Blue bin

B. Orange bin

C. Brown bin

2. Can used tissue paper go into the blue paper bin?

A. No: it is dirty and its fibres are too short

B. Yes: all paper can be recycled

3. What must be done with the fermented material from a bokashi bucket?

A. Use it straight away as potting soil for seedlings

B. Send it to the blue recycling bin

C. Bury it in soil or add it to a compost heap to finish breaking down

Answers: 1 A · 2 A · 3 C

🏠 Missions, scan & your town

✅ Home Missions

Real jobs to do at home, school or work: sort the family's recyclables, start a compost bottle, refuse single-use plastic for a week. Tick the steps, then finish the mission to upgrade the town. No photos are stored.

📷 Scan and sort

Point your camera at an item. WasteQuest makes a guess on your own device (nothing is uploaded), you correct it if needed, and it tells you which SWCorp bin to use: blue for paper, orange for plastic and metal, brown for glass. It never decides on hazardous waste.

🪙 Coins, shop & avatar

Right answers, badges and missions earn coins and XP. Spend coins in the Market shop on decorations for your town and accessories for your avatar.

Class Battle & Event Town

Class Battle: the teacher hosts a live quiz on the projector and students join with a code on their phones. No accounts are needed. Team mode works without phones.

Event Town: a whole crowd cleans one shared town on the big screen. Each right answer upgrades a district. Great for school events, recycling days and booths. A screen-only mode works without internet.

🏅 Badges & certificates

Collect badges from games, lessons and labs. Then pass the final quest for your track and print your own certificate of completion.



Junior Eco-Hero

Kids 7–12

10 questions from the kids track · pass mark 70% · earn at least 3 game badges



Zero-Waste Champion

Teens 13–17

15 questions from the teens track · pass mark 75% · at least 5 badges, including 1 lab badge



Waste-to-Wealth Practitioner

Adults

20 questions from the adults track · pass mark 80% · at least 2 lab badges + a W2W product pitch, verified by a teacher

These are non-credit certificates of completion issued by the WasteQuest programme.



Did you know?

Brown is for glass. Glass can be recycled again and again.

Part 4

Make: Maker Lab & Compost Farm

15 hands-on labs that turn real waste into useful things.





🦺 Golden safety rules

1. Read the whole lab before you start, and get everything ready.
2. ⛔ Steps marked ADULT (heat, hot oil, lye, sharp tools) are done by an adult only.
3. ⛔ Never melt or burn plastic: it gives off toxic fumes. Open burning is an offence in Malaysia.
4. Use clean, dry waste only. Never use medical waste, broken glass or chemical containers.
5. Never eat or taste anything in a lab, and wash your hands afterwards.
6. For a burn: cool it under cool running water for 20 minutes and tell an adult.

📋 Labs at a glance

Lab	Ages	⏱	🔥
🕯️ Eco-Candles from Used Cooking Oil	10+	60'	🔥
🐟 Fish-Waste Pet Treats	10+	150'	🔥
🎁 Trash to Treasure: Soap, Tote & Toy Car	8+	90'	🔥
🌈 Say No to Litmus: Rainbow pH from Kitchen Waste	7+	45'	–
☕ Kopi Wira Bau: Coffee-Ground Odour Absorbers	7+	40'	–
🌱 Self-Watering Bottle Garden	7+	40'	–
🍊 Eco-Enzyme	7+	30'	–
🧱 Eco-Bricks & Eco-Bags	7+	45'	–
🪱 Composting: Bokashi & Worms	7+	45'	–
🌽 Cornstarch Bioplastic	9+	60'	🔥
🪴 Bottle Vertical Garden	7+	60'	–

Lab	Ages	⏱	🔥
🥬 Bottle Hydroponics	9+	60'	–
☂️ Fused-Plastic Bags & Mini Umbrella	9+	90'	🔥
🛟 Bubble-Wrap Float Ring (Model)	7+	45'	–
🛌 3-in-1 Bubble-Wrap Sleeping Bag	10+	120'	–



Lab 1

Eco-Candles From Used Cooking Oil

Turn the oil left in the wok into a glowing scented candle.

Ages 10+ | ⏱ 60 min + 2-4 h to set | Difficulty ★★☆

Adult does hot or sharp steps | 🔥 Needs heat | SDG 6 | SDG 12 | SDG 13

🧒 **Age and supervision:** Primary pupils (up to 12): an adult does all the heating and pouring as a demonstration. Children filter cold oil, weigh and decorate.

Waste used
Used cooking oil

Product
Scented candles

Cost
Low: the oil is free. Example shop prices (Oct 2026): paraffin wax RM 14–22 per kg, so the 96 g in this recipe is about RM 1.30–2.10; wicks about RM 0.50 each; fragrance extra (one 10 mL bottle was RM 25). Prices vary by shop.



Scan to watch

🌏 Why it matters

🌍 Environment
Oil poured down the sink sticks inside pipes, causes blockages and pollutes rivers. Every jar of oil you reuse stays out of the drain.

💰 Economy
A free waste becomes a product people buy as gifts. Used cooking oil also has cash value: some recycling centres and collection points buy it for

🤝 Society
Candle-making is a simple home-business skill and a family activity that starts a conversation about how we throw away oil.

🧺 You need

- 48 g used cooking oil, cooled and left to settle
- 96 g paraffin wax (pellets or chips)
- Candle wicks with metal tabs, one per candle
- Moulds: small aluminium cups, clean tins or heat-proof glass jars
- Filter paper or a coffee filter, a funnel and a clean jar
- Kitchen scale
- Double boiler: a pot of water + a heat-proof jug or clean tin
- Wooden stick for stirring
- Candle dye or wax-crayon shavings (not food colouring)
- Candle fragrance oil or essential oil, a few drops
- Clothes pegs or chopsticks to hold the wicks straight
- Oven gloves, apron, a damp cloth and a pot lid close by

🪜 Steps

1. Ask an adult for cooled used cooking oil. Let it stand for a day so crumbs sink to the bottom.
2. Filter the oil through filter paper into a clean jar. Go slowly: it takes time!

   🔬 Food bits left in the oil do not burn cleanly: they char on the wick, smoke and smell.
3. Weigh 48 g of filtered oil and 96 g of paraffin wax. That is 1 part oil to 2 parts wax.

   🔬 Oil : wax = 48 : 96 = 1 : 2. Oil is liquid at room temperature, so too much oil gives a soft, greasy candle.
4. Stand a wick in the middle of each mould. Hold it straight with a clothes peg or two chopsticks across the top.

5 ADULT: put a few cm of water in a pot and heat gently. Stand the jug with the wax and oil in the water. Never put wax straight on the flame.

🔬 Water cannot get much hotter than 100 °C, so the wax stays far below the temperature at which it could catch fire.

6 ADULT: stir gently until everything is melted and clear.

7 ADULT: turn off the heat and lift the jug out with oven gloves. Children may drop in a little candle dye or crayon shavings from a spoon while the adult stirs.

8 ADULT: wait a minute, then add a few drops of fragrance oil and stir.

🔬 Fragrance evaporates from very hot wax; adding it just off the heat keeps more scent in the candle.

9 ADULT: pour the wax slowly into the moulds, keeping each wick in the centre.

10 Leave the candles for 2–4 hours until hard. Do not move them while they set.

⚠️ **The wax stays hot inside long after the top looks solid. Do not touch or poke it until it is hard and cool.**

11 Trim the wick to about 5 mm. Label your candle with the date and "made from used oil".

⚠️ **Test-burn one candle with an adult, on a heat-proof plate in a ventilated room. Note any smoke or smell.**

🦺 Safety First

⛔ Hot wax and hot oil cause serious burns. Only an adult heats and pours. Children stay an arm's length from the stove.

⛔ Double boiler only. Never heat wax directly over a flame, and never leave melting wax unattended.

⛔ If wax catches fire: turn off the heat and cover it with a pot lid. Never pour water on burning wax or oil.

⚠️ For a burn: cool it under cool running water for 20 minutes and tell an adult. Do not peel off wax stuck to the skin, and do not put butter, toothpaste or ice on a burn.

⚠️ Keep water away from hot wax and oil: even a few drops make hot fat spit and splash.

⚠️ Burn finished candles on a heat-proof plate, away from curtains. Never leave a lit candle alone.

🔬 The science Adults level

Fats, oils and grease (FOG) poured into drains solidify in sewers and cause blockages and overflows. Used cooking oil (UCO) has higher-value routes, especially collection for biodiesel, so a candle is best seen as a small upcycling and awareness product rather than the main solution. Be honest with buyers: paraffin is petroleum-derived (plant waxes can replace it at higher cost), and an oil-rich candle may smoke or smell slightly of cooking. Quality control (filtering, ratio, wick size) decides whether it is sellable.

👩‍🏫 **Teacher tip:** Filter the oil the day before: paper filtering is slow. One double-boiler station per adult. Pupils weigh, set wicks, choose colours and make labels; adults heat and pour.

🚀 Challenge

1. Make three mini-batches with oil : wax = 1:1, 1:2 and 1:3. Compare hardness, smoke and burn time. Which would you sell?
2. Burn rate: weigh a candle, burn it for 1 hour with an adult, weigh again. Burn rate = mass lost ÷ time (g/h). How many hours will a full candle last?
3. Costing: find real prices for wax, wicks and fragrance. Cost per candle = total cost ÷ number of candles. For a 40% margin, price = cost ÷ (1 − 0.40).

💭 Think about it

- Where does used cooking oil go in your home now?
- Why should we never pour oil down the sink?
- What would you change to make your candle better?
- Is a candle the best use for used oil? What else could it become?

✅ Quick quiz

1. Used cooking oil can be turned into...

A. Drinking water or juice

B. Glass jars or bottles

C. Candles, soap or biodiesel



Steps poster (BM). Photos and posters: UPM chemical engineering students, Faculty of Engineering.

2. What is the first step before making a candle from used cooking oil?

A. Filter out the food bits

B. Add cold water to it

C. Freeze the oil solid

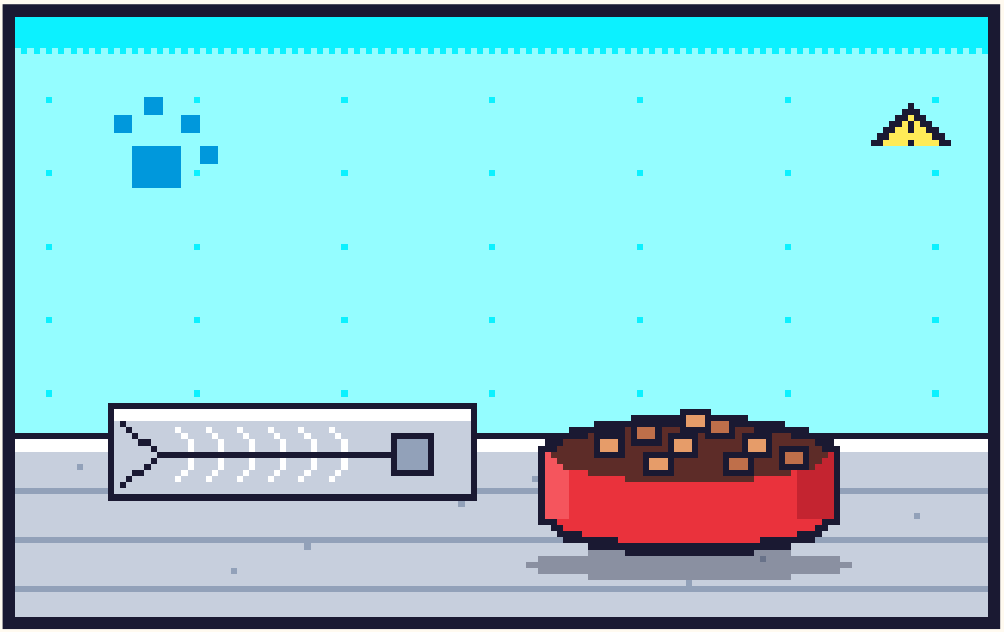
3. A candle recipe uses oil : wax = 1 : 2. You have 50 g of filtered oil. How much wax do you need?

A. 25 g

B. 50 g

C. 100 g

Answers: 1 C · 2 A · 3 C



Lab 2

🐟 Fish-Waste Pet Treats

Fish heads and scraps become crunchy baked treats for cats.

Ages 10+ | ⏱ About 2½ h (adult does the cooking) | Difficulty ★★★

Adult does hot or sharp steps | 🔥 Needs heat | SDG 12 | SDG 14

🧒 **Age and supervision:** An adult handles raw fish, the steamer and the oven. These are occasional treats, not a tested complete pet food.

Waste used
Fish trimmings: heads, skin, flesh scraps

Product
Baked cat treats

Cost
About RM 1.50 a batch, or RM 4.80 with taurine (2025 student prices).



Scan to watch

🌏 Why it matters

🌍 Environment

Fish heads and scraps are thrown away at markets and in kitchens, where they rot and smell. Food is the biggest part of Malaysian household waste (30.6%); using the scraps cuts it.

💰 Economy

Free market leftovers become a product pet owners pay for. The cost table shows where the money goes: the optional taurine powder is most of the cost.

🤝 Society

Community cat feeders and animal shelters need cheap, safe treats; this links food-waste reduction to animal welfare.

🧺 You need

- 283 g fresh fish trimmings (heads, skin, flesh scraps), used the same day and kept cold

- About 2 cups (283 g) wheat flour, plus a little for dusting
- 1 egg
- 1 tbsp (14 g) fresh cooking oil, not used oil
- Taurine powder: optional, only if a vet advises
- Steamer, oven, blender, kitchen scale, baking tray and baking paper
- Rolling pin, small cutters or a knife, bowls and spoons
- Gloves, a chopping board used only for raw fish, an airtight container

Ingredient	Amount	Cost (RM)
Fish waste	283 g	Free
Wheat flour	283 g	0.57
Egg	1 (50 g)	0.70
Fresh cooking oil	1 tbsp (14 g)	0.23
Taurine (optional)	11 g	3.30
Total		4.80 (1.50 without taurine)

Prices from the 2025 student poster. The poster's total of RM 5.03 was an adding mistake: the items add up to RM 4.80.

🪜 Steps

1. Wash your hands. Wearing gloves, rinse the fish trimmings well. ADULT removes the guts and gills with a knife (they spoil fastest).

   ⚠️ **Raw fish can carry bacteria: keep it cold and away from other food, and wash your hands with soap afterwards.**

2. ADULT: steam the fish for about 1 hour until fully cooked.

   🔬 Cooking kills germs and parasites and destroys thiaminase, an enzyme in some raw fish that breaks down vitamin B1.

3. Let it cool until it is comfortable to hold. Pick the flesh off and remove EVERY bone. Check twice with your fingers.
4. Blend the boneless fish into a smooth paste (purée). An adult fits, empties and washes the blender: the blades are sharp.
5. ADULT: spread the purée on a lined tray and dry it in the oven at about 117 °C for 15–20 minutes.
6. Mix the dried fish with the egg, flour and oil (and taurine only if a vet said so). Knead into a dough. No salt, onion, garlic or spices!

   ⚠️ **Onion and garlic are toxic to cats and dogs, even when cooked. The dough has raw egg in it: do not taste it, and wash your hands after kneading.**
7. Dust with flour, roll the dough flat and cut tiny shapes (cat-bite size).
8. ADULT: bake at 150 °C for about 20 minutes until firm and dry right through. Thicker pieces need longer.
9. Cool completely. Store in an airtight, labelled container in the fridge and use within about a week. Throw away anything that smells bad or grows mould.

   🔬 The 2025 student group saw no mould after 14 days, but home kitchens vary, so we play safe.
10. Give only a few pieces a day, as a treat.

    🔬 Vets advise that treats should be no more than 10% of a pet's daily calories.

🦺 Safety First

⛔ Steamer, oven, hot trays and knives: adults only.

⛔ Hygiene: wash hands, boards and knives after touching raw fish. Keep raw fish cold until cooking.

⚠️ This is a treat, not a complete diet, and not for humans. Ask a vet first, especially for kittens or pets with kidney problems or allergies.

⚠️ Remove all bones: small bones can choke a pet or cut its mouth.

⚠️ Check pupils' allergies (fish, egg, wheat) before the class.

🔬 The science Adults level

Fish processing leaves large volumes of by-products (heads, frames, skin, viscera). Valorisation routes include fishmeal, fish oil, fish silage, fertiliser and pet food. A home-scale treat demonstrates the idea, but selling pet food brings legal duties (hygiene, labelling, registration): check with the Department of Veterinary Services (DVS) before any commercial step. Nutritionally, a homemade treat cannot replace a complete and balanced diet.

🧑‍🏫 **Teacher tip:** If time is short, do steps 1–5 (raw fish, steaming, drying) before class; pupils mix, roll, cut and package. Use a separate board and knife for raw fish.

🚀 Challenge

1. Cost per treat: count the treats in one batch. Cost per treat = batch cost ÷ number of treats. Compare with a shop-bought cat treat (RM per 100 g).
2. Moisture loss: weigh the purée before and after drying. % water removed = (wet − dry) ÷ wet × 100.
3. Design a label: ingredients in order of weight, storage advice, "treat only, not for humans", and a use-by date.

💭 Think about it

- Which parts of a fish are usually thrown away at the market?
- Why do we dry and bake the treats instead of serving the fish paste?
- Why must the treats never contain onion, garlic or salt?
- Who in your community could use this idea?



Ingredients and steps 1–4 (BM). Photos and posters: UPM chemical engineering students, Faculty of Engineering.

✅ Quick quiz

1. Why should a home-made pet food from fish waste be cooked thoroughly (with an adult)?

- A. Cooking makes the food a nicer colour
- B. Heat kills harmful germs in raw fish waste
- C. Raw fish waste is too light to measure

2. Fish waste such as heads and bones can be turned into…

- A. Pet food or fertiliser
- B. Glass bottles or jars
- C. Batteries or wires

3. Biodiesel is made from used cooking oil by transesterification. The main by-product is…

- A. Petrol
- B. Glycerol
- C. Salt water

Answers: 1 B · 2 A · 3 B



Lab 3

🎁 Trash to Treasure: Soap, Tote & Toy Car

Three quick makes: used oil → soap, old T-shirt → bag, bottle caps → racing car.

Ages 8+ | ⏱ 3 activities × about 30 min | Difficulty ★★☆

Close supervision | 🔥 Needs heat | SDG 4 | SDG 6 | SDG 12

🧒 **Age and supervision:** Soap made with lye (sodium hydroxide) is for adults only. Children use melt-and-pour soap base, with an adult doing the melting.

Waste used
Used cooking oil, old T-shirts, bottle caps, ice-cream sticks

Product
Soap, tote bag, toy car

Cost
Very low: mostly waste. Example shop price (Oct 2026): melt-and-pour soap base RM 26 per kg, so 200 g is about RM 5.20. Glue, rubber bands and decorations extra. Prices vary.



Scan to watch

🌏 Why it matters

🌍 Environment
Textiles are 2.3% of Malaysian household waste, and oil in drains pollutes rivers. Reusing a T-shirt, oil or bottle caps keeps

💰 Economy
Upcycled soap and bags sell at school bazaars; toys from waste cost almost nothing.

🤝 Society
Making things builds creativity, problem-solving and pride. Old T-shirts carry memories: "every bag tells a story".

🧺 You need

Activity A: soap

- Children: 200 g glycerin melt-and-pour soap base (no lye)
- Optional: 1 tsp dried used coffee grounds (scrub) and 1–2 drops essential oil
- Moulds from waste: yogurt cups, cut milk cartons
- Teacher demo only: 500 g filtered used palm cooking oil, 65 g sodium hydroxide (NaOH), 160 g cold distilled water, goggles, chemical gloves, long sleeves, stainless-steel or heat-proof PP plastic jugs, digital scale, stick blender, lined mould

Activity B: T-shirt tote

- 1 old T-shirt (thick cotton works best)
- Sharp fabric scissors, a ruler and chalk or a marker
- Optional: fabric paint or fabric markers

Activity C: toy car

- 4 bottle caps of the same size
- 2–3 ice-cream sticks and 1 drinking straw
- 2 wooden skewers or long toothpicks (axles)
- 1 rubber band and some glue

🪜 Steps

tivity A: soap

1. CHILDREN: an adult cuts the soap base into cubes and melts it in short bursts in a microwave, or in a double boiler.
2. Stir in a pinch of dried coffee grounds and 1–2 drops of essential oil.

   ⚠️ **Melted soap base is hot (about 60–70 °C): the adult holds the jug steady while you stir with a long spoon.**

3. ADULT pours it into the moulds. After about 1 hour, pop the soap out. Ready to use!
4. TEACHER DEMO (adults only, full PPE): weigh 160 g cold distilled water into a heat-proof PP or stainless jug. Outdoors or by an open window, slowly add 65 g NaOH TO the water (never water to lye), stirring. It gets very hot and gives off fumes: stand back.

   🔬 Lye amount: 500 g × SAP value 0.138 (palm olein) ≈ 69 g; minus about 5% "superfat" so a little oil stays unreacted ≈ 65 g. Always re-check your actual oil in a lye calculator.
5. TEACHER DEMO: when the lye solution and the filtered oil have both cooled to warm (about 40 °C), pour the lye into the oil and stick-blend until it thickens like custard ("trace"). Pour into the lined mould.
6. TEACHER DEMO: cover and leave 24–48 h, unmould with gloves, then cure in an airy place for 4–6 weeks. Fresh lye soap is caustic: do not touch it or take it home until cured. Use UCO soap for laundry and cleaning, not skin.

   🔬 Test the cured soap in the Litmus lab: bar soap is naturally alkaline, about pH 9–10.

tivity B: T-shirt tote

7. Lay the T-shirt flat. Cut off both sleeves just inside the seams.
8. Cut a big scoop around the neck to make the bag opening. The shoulder parts become the handles.
9. Turn the shirt inside out. Draw a line about 8 cm up from the bottom hem.
10. Cut strips from the bottom up to the line, about 2–3 cm wide, through both layers.

11 Tie each front strip to the back strip under it with a double knot.

🔬 Then tie one strand of each pair to its neighbour's strand to close the small gaps.

12 Turn the bag right side out: the knots are hidden inside. Decorate it!

tivity C: toy car

13 Cut the straw into two pieces, a little wider than an ice-cream stick. Glue them across the stick, one near each end. These hold the axles.

14 ADULT: make a small hole in the centre of each bottle cap with a bradawl or small drill.

15 Snip the sharp points off the skewers first (ADULT, with strong scissors). Push a skewer through one straw and fix a cap on each end with a dab of glue. Repeat at the other end.

16 Loop the rubber band over the front of the car and hook the other end around the back axle.

17 Roll the car backwards to wind the band round the axle, put it down and let go. Vroom!

🔬 Test, then improve: does it go straight? Do the wheels slip? A thin rubber-band "tyre" on the back wheels adds grip.

🦺 Safety First

⛔ Lye (sodium hydroxide, NaOH) is corrosive: it burns skin and can blind. Teacher demo only, with goggles, chemical gloves, long sleeves and good ventilation; children watch from at least 2 metres away. Add lye to water, never water to lye. Never use aluminium containers (they react and make hydrogen gas).

⛔ NaOH looks like sugar or salt. Keep it in its original, labelled container, locked away from children, and never in a food or drink container.

⛔ If lye touches skin or eyes: rinse with plenty of running water for at least 15 minutes (water only: no vinegar). For eyes, get medical help immediately.

⛔ Melted soap base is hot: an adult melts and pours it.

⚠️ Sharp tools: an adult makes the bottle-cap holes. Cut away from your body.

⚠️ Bottle caps are a choking hazard: keep them away from children under 3.

🔬 The science Adults level

The trio shows three circular strategies: chemical conversion (UCO → soap), reuse and repurposing (T-shirt → bag, with no reprocessing energy) and creative reuse of low-value plastics (caps). Cold-process soap is a chemical process for trained adults; melt-and-pour keeps the session child-safe. Commercially, UCO soap is usually positioned as laundry or cleaning soap; soap sold for use on skin is treated as a cosmetic in Malaysia and needs notification with the NPRA, so check the rules before selling.

🧑‍🏫 **Teacher tip:** Run the three activities as stations. Do the lye demo only if you are trained; otherwise do melt-and-pour. Collect T-shirts and bottle caps a week ahead.

🚀 Challenge

1. Lye maths: with SAP = 0.138 g NaOH per g oil and 5% superfat, how much NaOH does 300 g of oil need? (300 × 0.138 × 0.95 ≈ 39 g)
2. Car race: measure the distance travelled for 1, 2 and 3 winds of the axle. Plot distance against winds. Is it a straight line?
3. Market test: price a T-shirt tote for a school bazaar. What is your cost (materials + time)? What would buyers pay?

💭 Think about it

- Which activity saved the most waste? How do you know?
- Why is lye soap made only by trained adults?
- What other old clothes or toys could you upcycle?

✅ Quick quiz

1. Making soap from oil uses lye (sodium hydroxide, NaOH). What is the key safety rule?

A. Taste a tiny drop to check its strength

B. Stir it quickly with bare hands to save time

C. Adult supervision, gloves and goggles: lye burns skin



Activity 1: used oil to soap. Photos and posters: UPM chemical engineering students, Faculty of Engineering.

2. The reaction of oil or fat with sodium hydroxide to make soap is called...

A. Photosynthesis

B. Saponification

C. Fermentation

D. Combustion

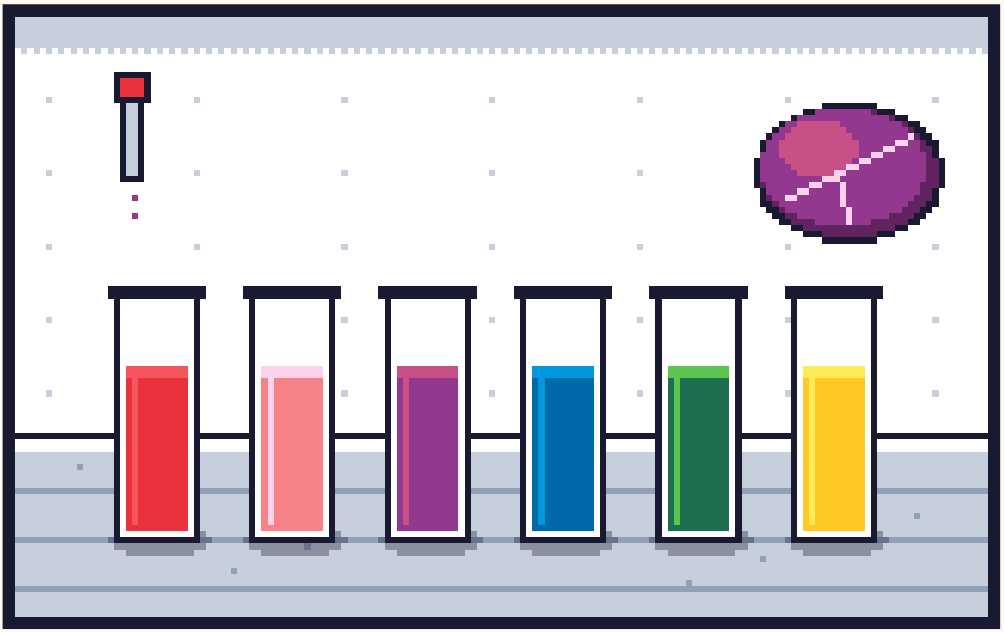
3. A W2W social enterprise that pays fair wages to local women making soap from used oil links most directly to...

A. SDG 14 Life below water

B. SDG 6 Clean water and sanitation

C. SDG 8 Decent work and economic growth

Answers: 1 C · 2 B · 3 C



Lab 4

🌈 Say No to Litmus: Rainbow pH From Kitchen Waste

Make your own pH indicator from cabbage leaves or bunga telang: no bought litmus paper needed.

Ages 7+ | ⏱ 48 min | Difficulty ★☆☆ | Close supervision | 🌿 No heat

SDG 4 | SDG 6 | SDG 14

Waste used
Purple-cabbage outer leaves, wilted bunga telang flowers, scrap paper

Product
Natural pH indicator and test strips

Cost
Almost free: household items.



Scan to watch

🌏 Why it matters

🌍 Environment

Strong acids or bases poured down drains change the pH of rivers and harm fish and useful bacteria. Knowing pH helps us dispose of liquids safely.

💰 Economy

A free indicator from kitchen waste replaces bought litmus paper, saving school lab money.

🤝 Society

Testing household liquids builds science confidence and helps families think about what they pour away.

🧺 You need

- 3–4 purple-cabbage outer leaves, OR about 10 bunga telang (butterfly-pea) flowers

- A blender, or a bowl and spoon to squash; a sieve or clean cloth
- 6–8 clear cups (or bottles cut in half), labelled
- Samples: lime juice, vinegar, soda water, tap water, salt water, baking-soda water (1 tsp in ½ cup water), soapy water
- Spoons or droppers (washed medicine syringes work)
- Scrap paper or coffee filters for test strips
- Teacher: a pH meter or universal indicator paper to check
- Safety glasses, if available

🪜 Steps

1. Tear the cabbage leaves (or flowers) into small bits. Squash them with 1 cup of water for 2 minutes, or an adult runs the blender.
   🔬 Warm water, added by an adult, pulls the colour out faster.
2. Strain the liquid through a sieve or cloth. You now have a purple (cabbage) or blue (bunga telang) indicator!
3. Pour a little of each sample into its own labelled cup.
4. Guess first! Write "acid", "neutral" or "base" for each cup.
5. Add 2 spoons of indicator to each cup. Watch the colour change and write it down.
6. Line the cups up from red or pink (acid) through purple or blue (neutral) to green or yellow (base). You made a pH rainbow!
   🔬 Roughly, red cabbage goes red (pH 1–2) → pink (3–4) → purple (5–7) → blue (8) → blue-green (9–10) → green (11–12) → yellow (13+). Bunga telang is pink-purple in acid, blue near neutral and green in base.
7. Precision guess (the poster's game): the teacher hides the pH meter's screen. Guess each sample's pH number, then reveal. Who was closest?

8. Mix it right! Put 1 spoon of vinegar and 2 spoons of indicator in a cup. Add baking-soda water a little at a time (a spoon or 1 mL from a syringe), stirring, until the colour turns back to purple or blue (neutral). Count how much you added. Watch it fizz!
   🔬 The fizz is carbon dioxide: $CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + H_2O + CO_2$.
9. Bonus: soak strips of scrap paper or coffee filter in the indicator and let them dry. Now you have your own "litmus" paper!
10. Clean up: these household samples can go down the sink with plenty of water. Leftover cabbage goes to the compost.

🦺 Safety First

⚠️ Never taste or touch the test liquids: we look at the colour instead. (The old poster said acids "taste sour" and bases "feel slippery"; we never test that way.)

⚠️ Lime juice and vinegar sting the eyes. Wear glasses if you have them, and rinse with clean water if splashed.

⚠️ Household samples only: no bleach, drain cleaner or toilet cleaner. Never mix cleaning products.

🔬 The science Adults level

pH is a core parameter in waste management: landfill leachate, food-waste fermentation (eco-enzyme is about pH 3.5–4) and industrial effluent are all monitored for pH, and Malaysian effluent regulations set allowed pH ranges for discharges. A zero-cost indicator made from kitchen waste lets community groups screen liquids and teach the concept before investing in meters.

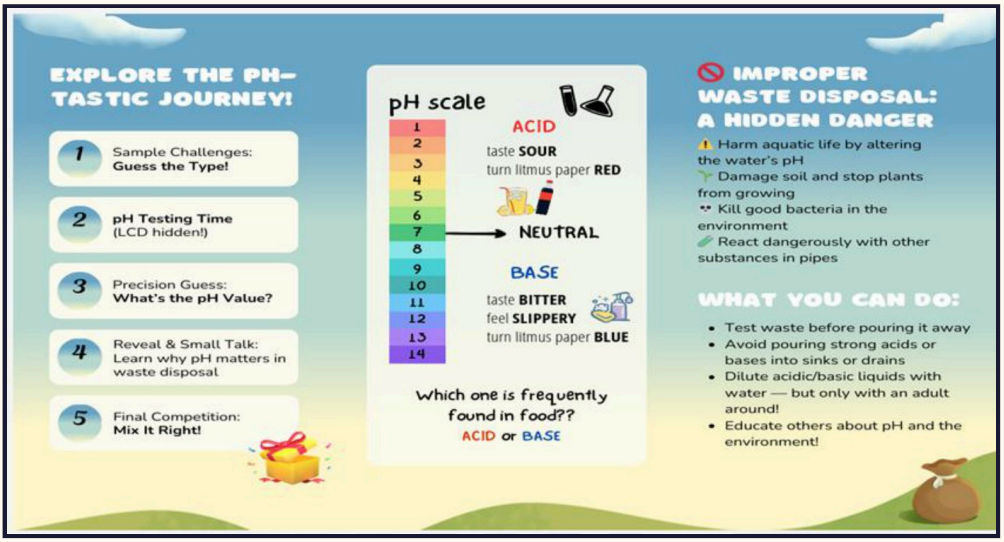
👩‍🏫 **Teacher tip:** Make one big jug of indicator before class. Use the hidden-meter game with older pupils; for under-9s, stop at the colour rainbow. Link it to other labs: test the eco-enzyme and the cured soap.

🚀 Challenge

1. Colour chart: measure samples with a pH meter and photograph the indicator colour at each pH value.
2. Cabbage vs bunga telang: which shows more colour steps? Which keeps its colour longer?
3. Titration: how many mL of baking-soda solution (from a syringe) neutralise 5 mL of vinegar? Repeat 3 times and average. Predict, then test, what happens with 10 mL. (Expect a lot: household vinegar is much stronger than the baking-soda water.)

💭 Think about it

- Which sample surprised you most?
- Are most foods acidic or basic? Look at your results!
- Why is it dangerous to pour strong acids or bases down the drain?
- How did a "waste" cabbage leaf replace something we usually buy?



pH journey poster. Photos and posters: UPM chemical engineering students, Faculty of Engineering.

✅ Quick quiz

1. Why make pH indicators from kitchen waste (e.g. cabbage leaves, flower petals) in class?

A. They are cheap, safe and reuse waste: a W2W science tool

B. They are more precise than a pH meter

C. They only work at night

2. A pH of 7 means the liquid is…

A. Neutral

B. Very acidic

C. Very alkaline

3. Which natural pigment makes red cabbage and butterfly pea change colour with pH?

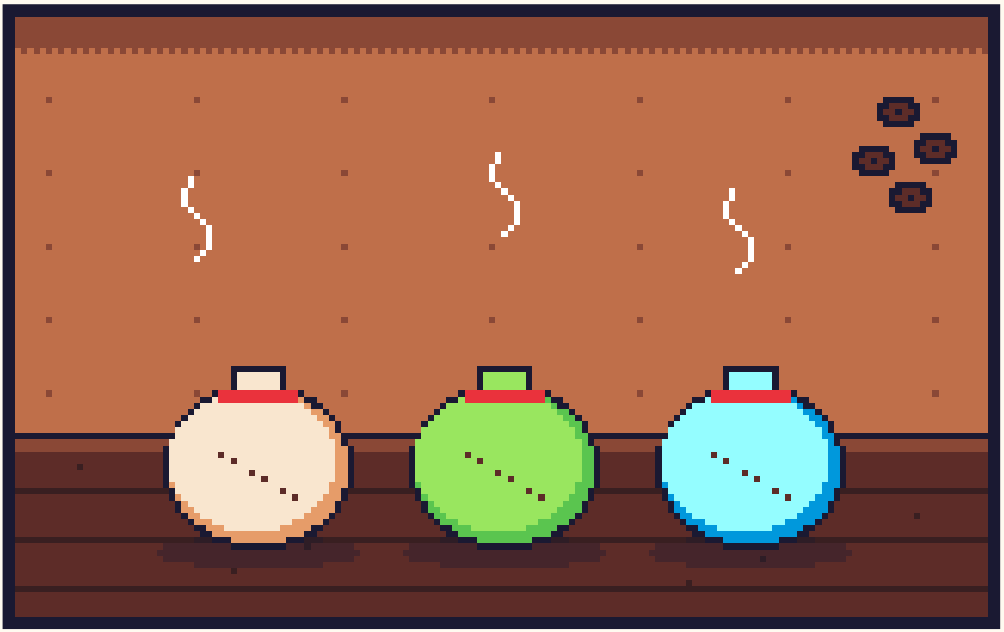
A. Chlorophyll

B. Starch

C. Anthocyanins

D. Protein

Answers: 1 A · 2 A · 3 C



Lab 8

☕ Kopi Wira Bau: Coffee-Ground Odour Absorbers

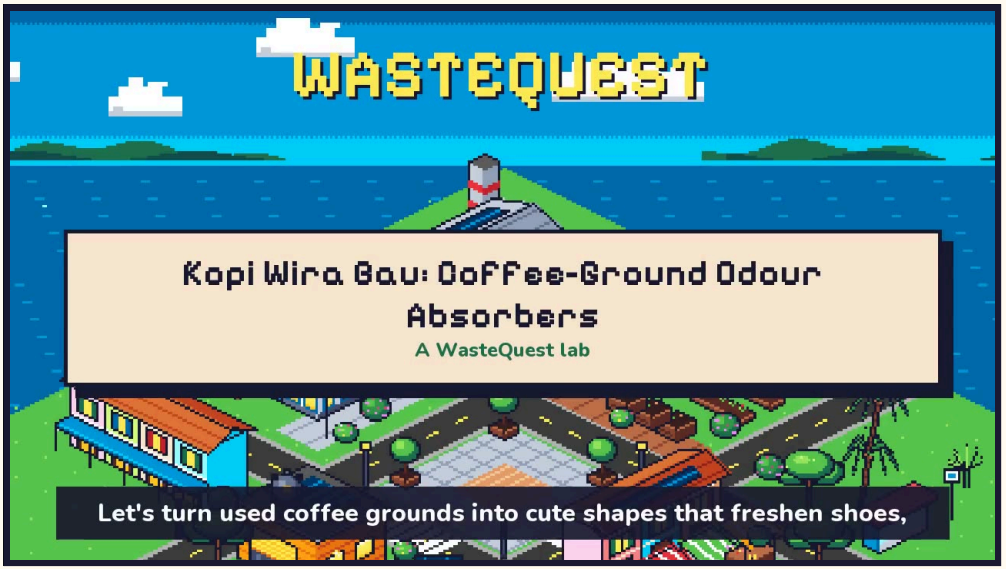
Used coffee grounds become cute shapes that freshen shoes, cupboards and fridges.

Ages 7+ | ⏱ 40 min + 1-2 days drying | Difficulty ★☆☆ | Light supervision | 🌿 No heat | SDG 11 | SDG 12

Waste used
Used coffee grounds from cafés

Product
Odour-absorbing shapes

Cost
Very low: the cornstarch and corn syrup for one batch cost about RM 0.90–1.00 (shop prices, Oct 2026). Add a little for cinnamon and fragrance. Prices vary.



Scan to watch

🌏 Why it matters

🌍 Environment

Cafés throw away coffee grounds every day; in landfill they rot. Reusing them keeps food waste out of the bin.

💰 Economy

A free waste becomes a gift or bazaar product that is cheaper than many shop-bought air fresheners.

🤝 Society

Pupils ask a local café for its grounds: a small partnership between school and business.

🧺 You need

- 1 cup used coffee grounds, completely dried
- ½ cup cornstarch
- 2 tbsp corn syrup (the binder)
- 1 tsp ground cinnamon

- A little water, only if needed
- Tray, bowl and spoon
- Silicone moulds (chocolate or ice-cube shapes)
- A small open container or cloth pouch (an old sock works!)

🪜 Steps

1. Ask a nearby coffee shop for its used coffee grounds. Bring a clean container.
2. Spread the grounds thinly on a tray. Dry them for 1–2 days in the sun or an airy place, stirring sometimes. They must be completely dry.

   ⚠️ **Damp grounds go mouldy within a few days. If you see fuzzy mould, do not sniff it: throw that batch in the compost and wash your hands.**
3. In a bowl, mix 1 cup dry grounds, ½ cup cornstarch and 1 tsp cinnamon.
4. Add 2 tbsp corn syrup and the drops of fragrance (adult). Mix well with a spoon.
5. Squeeze a little in your hand. If it crumbles, add water one teaspoon at a time until it holds together.
6. Press the mixture firmly into the silicone moulds.
7. Dry for 24–48 hours until hard, then pop them out.
8. Put them in a small open container or cloth pouch: shoe rack, wardrobe, fridge or bathroom.
9. When the smell fades, dry them in the sun to refresh. Old ones can go into the compost.

🦺 Safety First

⚠️ Keep away from pets and toddlers: coffee (caffeine) and essential oils are harmful to dogs and cats. Label them "Not food": they look like chocolate!

⚠️ Essential oils can irritate skin: an adult adds the drops. Wash hands afterwards.

⚠️ Throw away any shape that grows mould. The sugary binder can attract ants: keep shapes in a closed container with small holes, not loose on the floor.

⚠️ In the fridge, use an unscented shape (no fragrance oil) so the food does not pick up the smell.

🔬 The science Adults level

Spent coffee grounds are a steady, clean, single-source waste stream from cafés, ideal for small W2W products. Other proven uses are compost, mushroom-growing substrate and fuel pellets. Keep product claims precise: "helps reduce and mask odours" is honest; "neutralises odours" or "antibacterial" would need testing.

👩‍🏫 **Teacher tip:** Collect and dry the grounds a week before class: drying decides success. One batch is enough for a small group; scale up by cups.

🚀 Challenge

1. Fair test: put half a cut onion in two closed boxes, one with a coffee shape and one without (try baking soda too). After a day, blind testers rate the smell from 1 to 5.
2. Drying: weigh the wet grounds and the dry grounds. What percentage was water?
3. Price it: cost per shape, then a selling price for a pack of 3 including packaging.

💭 Think about it

- Where else could you use your Kopi Wira Bau?
- Why must the grounds be completely dry?
- Which other café wastes could become a product?

✅ Quick quiz

1. Dried used coffee grounds can be made into…

A. An odour absorber

B. Drinking water

C. Glass



Materials. Photos and posters: UPM chemical engineering students, Faculty of Engineering.

2. A compost pile smells of rotten eggs. What is the most likely cause and fix?

A. Too wet, not enough air: add browns and turn it

B. Too dry, too much air: add water and sand

C. Too cold: cover it tightly with a plastic sheet

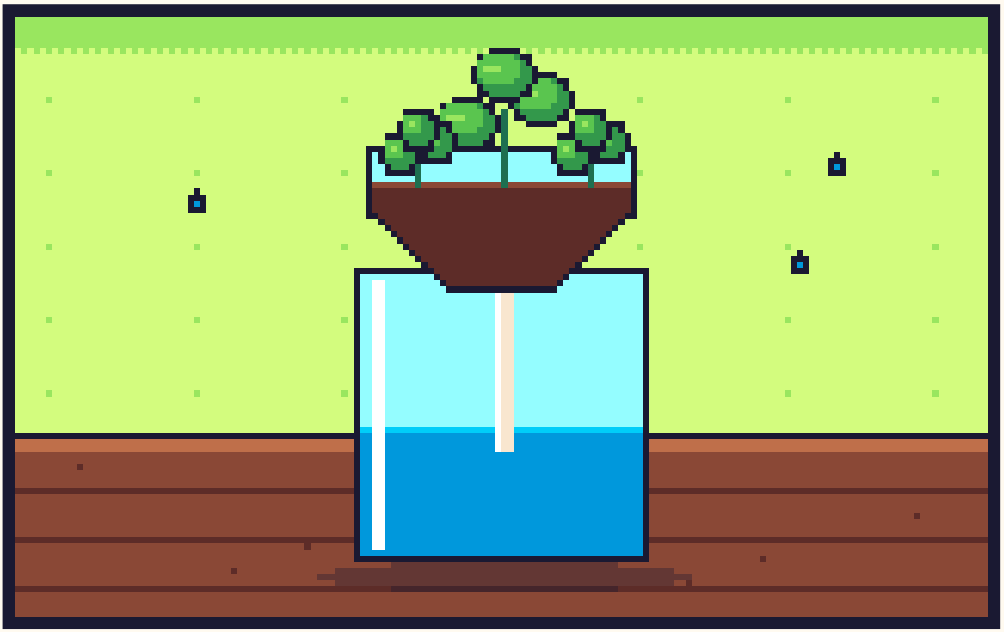
3. Finished compost looks and smells…

A. Bright and shiny

B. Dark, crumbly and earthy

C. Like rotten eggs

Answers: 1 A · 2 A · 3 B



Lab 6

🌱 Self-Watering Bottle Garden

A plastic bottle and a T-shirt strip water your plant for you. Teens can add an Arduino brain.

Ages 7+ | ⏱ 40 min (Arduino version: +2 h) | Difficulty ★★☆

Close supervision | 🌿 No heat | SDG 4 | SDG 9 | SDG 13 | SDG 18

🧒 **Age and supervision:** Arduino version: low-voltage parts only (USB or batteries), never mains electricity. An adult checks the wiring before power goes on.

Waste used
PET bottles, old T-shirt

Product
Self-watering planter (automatic watering for teens)

Cost
Wick version: almost free. Arduino version: example parts (Oct 2026: compatible board, moisture sensor, relay, small pump) add up to about RM 32–57, before wires, tubing and a power supply. Prices vary by shop.



Scan to watch

🌏 Why it matters

🌍 Environment
Bottles are reused before they are recycled, and wick watering gives the plant only the water it needs, with less run-

💰 Economy
Free planters; automatic watering saves time and water for urban growers.

🤝 Society
Small gardens fit flats and school corridors, adding green space where there is little (an idea from a 2024 student project).

🧺 You need

Wick planter (everyone)

- 1 clean 1.5 L PET bottle, label removed
- 1 strip of old cotton T-shirt, about 2 cm × 25 cm (the wick)
- Potting soil mixed with some compost
- Seeds or seedlings: kangkung, sawi, bayam or chilli
- Scissors or craft knife and a nail or bradawl (adult)
- Paint or markers to decorate

Automatic version (teens)

- Arduino Uno or Nano + USB cable or 5 V power bank
- Capacitive soil-moisture sensor (not a DHT11: that one measures air)
- 5 V relay module
- Small 3–5 V submersible pump with its own battery pack + 1 m silicone tube
- Jumper wires and a covered water container

🪜 Steps

ick planter

1. ADULT: cut the bottle in half, a little below the middle.
2. ADULT: make a hole in the bottle cap with a nail or bradawl.
3. Push the T-shirt strip through the cap hole and tie a knot so it can't slip out. About 10 cm hangs below the cap.
4. Wet the wick. Screw the cap on and turn the top half upside down (cap pointing down) into the bottom half.
5. Fill the top part with moist soil, spreading the wick inside the soil. Plant your seeds.

6. Lift the top out, pour water into the bottom part, and put the top back. The wick must touch the water.
7. Decorate your planter and put it where it gets light. Check the water level every few days.
8. Every week: empty the reservoir, rinse it and refill. This stops Aedes mosquitoes breeding!

tomatic version (teens)

9. Wire it: sensor VCC → 5V, GND → GND, AOUT → A0. Relay IN → D7, VCC → 5V, GND → GND. The pump's battery pack is switched through the relay's COM and NO terminals.
10. Calibrate: open the Serial Monitor and note the reading in dry air and with the sensor tip in a glass of water. Set THRESHOLD between the two.

    🔬 Most capacitive modules read higher when dry and lower when wet.
11. Upload the sketch below. Push the sensor into the soil (only up to its line) and test: dry soil → pump runs 2 seconds → waits → reads again.
12. Keep all electronics in a box above the water and log how often it waters.

💻 Arduino sketch

🦺 Safety First

⚠️ Cut bottle edges can be sharp: an adult cuts, and you can cover the edge with tape.

⛔ Electricity and water (Arduino version): use only low voltage (USB, power bank or AA batteries). Never use mains (230 V) power. Keep the electronics above and away from the water, touch wires only with dry hands, and never let the pump run dry.

⚠️ Standing water breeds Aedes mosquitoes (dengue). Keep the reservoir covered and change the water at least once a week: mosquitoes grow from egg to adult in about 7–10 days.

🔬 The science Adults level

Sub-irrigation (wicking) cuts evaporation and run-off and suits small urban plots; sensor-driven irrigation adds data and saves labour. Costs scale quickly: one 2024 student proposal budgeted RM 51 per pot for a sensor-and-pump vertical garden, so the wick system is the low-cost entry point and automation suits demonstrations or high-value crops. In the waste hierarchy, reusing a bottle comes before recycling it.

👩‍🏫 **Teacher tip:** Cut the bottles before class for under-10s. Fast growers (kangkung, sawi) show results within about 2 weeks. Run the Arduino build as a teen club activity.

🚀 Challenge

1. Water use: mark the reservoir level every day. How many mL does the plant use per day? Compare a wick pot with a hand-watered pot.
2. Wick test: try cotton, polyester and a sponge strip. How high does coloured water rise in 10 minutes in each?
3. Arduino: print a reading every 10 minutes and graph soil moisture over 3 days. Add a daily limit on pump time to the code.

💭 Think about it

- How does the water get from the bottle to the roots without you?
- Why must we change the water every week?
- What else could you grow in a bottle garden at school?



How the automatic system works, benefits. Photos and posters: UPM chemical engineering students, Faculty of Engineering.

✅ Quick quiz

1. SDG 12 is about...

- A. Responsible consumption and production
- B. Clean water and sanitation
- C. Industry, innovation and infrastructure

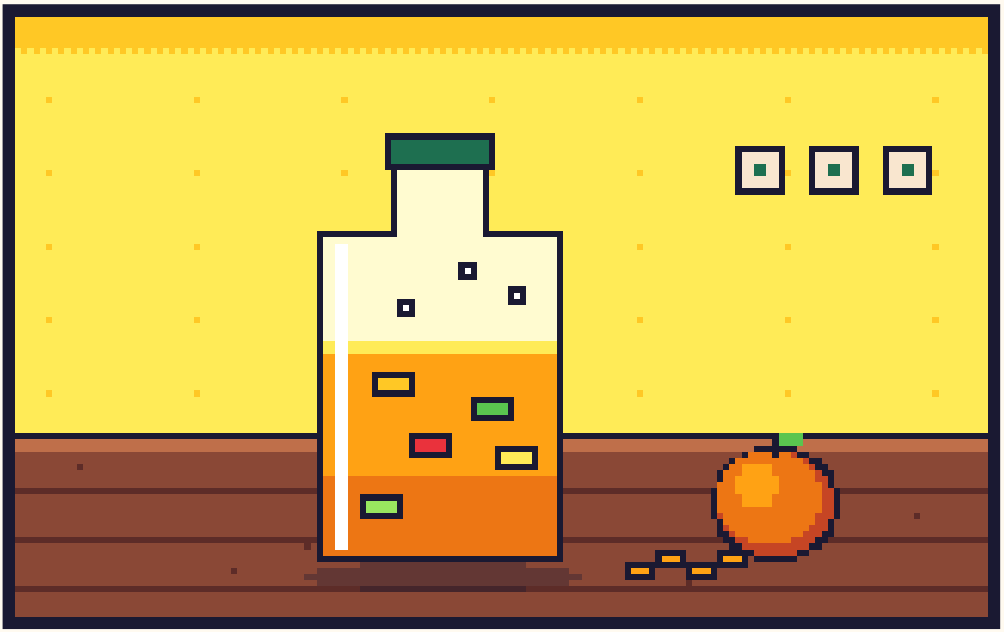
2. SDG 13 is...

- A. Quality education
- B. Climate action
- C. Clean water and sanitation

3. SDG 15 is...

- A. Life below water
- B. Affordable and clean energy
- C. Life on land

Answers: 1 A · 2 B · 3 C



Lab 7

🍊 Eco-Enzyme

Fruit peels + brown sugar + water and a little patience make a natural cleaner.

Ages 7+ | ⏱ 30 min + 3 months fermenting | Difficulty ★☆☆

Light supervision | 🌿 No heat | SDG 6 | SDG 12 | SDG 13

Waste used
Fruit and vegetable peels

Product
Eco-enzyme liquid (mild cleaner, plant feed)

Cost
Very low: about RM 0.60 of brown sugar for a batch made with 1 L of water (shop prices, Oct 2026). How much liquid you get back varies.



Scan to watch

🌏 Why it matters

🌍 Environment
Food is 30.6% of Malaysian household waste. Peels turned into eco-enzyme stay out of landfill, where they would rot and release methane.

💰 Economy
A bag of sugar plus free peels makes litres of mild cleaner, and the leftover pulp becomes compost.

🤝 Society
Easy for every home and school; eco-enzyme groups are active in many Malaysian communities.

🧺 You need

- 100 g brown sugar, gula merah or molasses (1 part)
- 300 g fresh fruit and vegetable peels, cut small (3 parts): pineapple, orange, papaya, banana…

- A 2 L plastic bottle or container with a screw lid (not glass). The mix fills about 1.4 L, so a 1.5 L bottle is too full.
- Kitchen scale, knife, chopping board and funnel
- Label and marker

🪜 Steps

1. Wash and chop the peels into small pieces (children use kitchen scissors; an adult cuts hard peels such as pineapple with a knife). No meat, oily food or mouldy fruit.
2. Weigh 1 part sugar, 3 parts peels and 10 parts water: 100 g sugar, 300 g peels, 1,000 g (1 L) water.

   🔬 The ratio is by weight. The 2025 poster scaled it up to 1 kg : 3 kg : 10 kg.
3. Pour the water into the bottle, add the sugar and shake until it dissolves.
4. Add the peels. Leave at least a quarter of the bottle empty for the gas.
5. Close the lid. Label it with today's date and the ready date (3 months later).
6. Keep it in a cool, shady place, out of direct sun.
7. For the first 2 weeks, open the lid a little every day to let the gas out, then close it. After that, once a week. Children: ask an adult to open it, slowly, pointing away from faces.

   🔬 The 2025 student group built a pressure-relief valve and an Arduino-driven stirrer to do this automatically.
8. After 3 months, strain it. The brown liquid is eco-enzyme! Put the leftover pulp in compost or in your next batch.

9 Always dilute before use. Commonly used: about 1 part eco-enzyme to 10–20 parts water for wiping floors and tables, and much weaker (about 1 : 500–1,000) for watering plants.

🦺 Safety First

⚠️ Gas builds up inside: release it daily at first or the bottle can bulge or burst. Never use glass bottles. If the bottle feels hard, open it very slowly over a sink, pointing away from faces: it can spray.

⚠️ Keep it out of hot places such as a car or a sunny window: heat makes the gas build up faster.

⚠️ Not for drinking. Keep it labelled and away from small children. It is acidic: rinse eyes with water if splashed.

⚠️ A thin white film on top is normal. If you see black mould or it smells rotten, add a little more sugar, close it and check again in a week; if it is still bad, compost it and start again.

🔬 The science Adults level

Eco-enzyme is a low-cost, low-tech way to divert fruit waste and engage communities. Be careful with claims: the name suggests strong enzyme activity, but its cleaning action comes mainly from organic acids. It is not a registered disinfectant, and claims that it purifies rivers or treats illness are not supported by solid evidence. Honest positioning (a mild acidic cleaner and a way to cut food waste) protects your credibility.

👩‍🏫 **Teacher tip:** Start a class batch early in the term so it is ready before the end. Set a rota for the daily gas release. Test the pH every month in the Litmus lab.

🚀 Challenge

1. Measure the pH every 2 weeks for 3 months and plot it. When does it level off?
2. Fair test: one bottle with pineapple peel, one with banana peel. Compare smell, colour and final pH.
3. Cost per litre: price of the sugar ÷ litres made. Compare with a shop floor cleaner (per litre of diluted cleaner).

💭 Think about it

- What made the gas inside the bottle?
- Why do we leave space at the top?
- Which claims about eco-enzyme are true, and which need more evidence?

✅ Quick quiz

1. How long does eco-enzyme usually ferment?

A. About 3 months

B. About 3 hours

C. About 3 days



Materials: eco-enzyme, soap and mixer. Photos and posters: UPM chemical engineering students, Faculty of Engineering.

2. Why do we open the eco-enzyme lid a little in the first weeks (with an adult)?

A. To let flies in to help it ferment

B. To let out gas so it does not burst

C. To add sand to make it heavier

3. Which container is best for fermenting eco-enzyme?

A. A glass bottle filled to the brim and sealed tight

B. An open bowl

C. A plastic container with space left at the top

Answers: 1 A · 2 B · 3 C



Lab 8

Eco-Bricks & Eco-Bags

Pack soft plastic into a bottle-brick, and weave snack wrappers into a tough bag.

Ages 7+ | ⏱ 45 min a session (one brick takes several sessions)
Difficulty ★☆☆ | Light supervision | 🌿 No heat | SDG 11 | SDG 12 | SDG 13 | SDG 14 | SDG 15

Waste used

Clean, dry soft plastics; PET bottles; foil-lined snack wrappers

Product

Eco-bricks (benches, garden borders) and a woven wrapper tote

Cost

Free: all waste, plus staples or tape.



Scan to watch

🌏 Why it matters

🌍 Environment

Plastic is 21.9% of Malaysian household waste, and soft plastics like wrappers are hard to recycle. Packed into eco-bricks, they stay out of drains, rivers and open burning.

💰 Economy

Eco-bricks replace some bought materials for garden benches and borders; woven bags are durable, waterproof and unique.

🤝 Society

Building a school bench together is a visible shared achievement (one 2024 student team gave bags to homeless people).

🧺 You need

Eco-brick

- 1 PET bottle (one size for the whole class, e.g. 600 ml or 1.5 L)

- Soft plastics: wrappers, plastic bags, straws, all clean and completely dry
- A packing stick (wooden spoon handle or bamboo stick)
- Scissors and a kitchen scale

Eco-bag (wrapper weave)

- Lots of foil-lined snack or coffee wrappers, washed and dried
- Ruler, pencil and scissors
- Stapler or tape, and 2 long strips for handles

Bottle	Minimum mass	Density
500 mL	167 g	0.33 g/mL
600 mL	200 g	0.33 g/mL
1.5 L	500 g	0.33 g/mL

Global Ecobrick Alliance minimum for building: 0.33 g/mL.

🪜 Steps

o-brick

1. Wash and dry the plastics. No food, paper, glass, metal or anything sharp.
2. Cut big pieces smaller. Push one coloured piece into the bottom first: it looks nice when you build.
3. Add a little plastic at a time and press it down hard with the stick. Keep packing!
4. Keep going until you can't push any more in. Squeeze the bottle: it should feel like a brick, not a balloon.

5 Weigh it. Target: at least 0.33 g for every mL of bottle: 200 g for 600 mL, 500 g for 1.5 L.

🔬 Density = mass ÷ volume. 200 g ÷ 600 mL = 0.33 g/mL. Lighter bricks are too squishy to build with.

6 Close the cap and write the mass and date on the bottle.

o-bag (wrapper weave)

7 Cut the wrappers into rectangles 16 cm × 6 cm.

8 Fold each rectangle in half the long way, open it, fold both long edges to the middle line, then fold in half again. You get a long thin strip.

9 Fold the strip in half across the middle, then fold both ends in to the middle crease. You get a small V-shaped link.

10 Slide the ends of one link into the pockets of another to make a zig-zag chain. Make many chains.

11 Join chains side by side with staples or tape to make two panels, then join the panels into a bag and add two handles.

🔬 One 2024 student team filled small gaps with thin plastic strips. Put shiny sides out for a sparkly bag.

🦺 Safety First

⚠️ Clean and dry only: food leftovers make bricks mouldy and smelly.

⚠️ Scissors: cut away from your body. Staples are sharp.

⚠️ Eco-bricks are not for load-bearing walls. Keep them out of strong sun and away from fire: use them inside a frame, or covered with earth or cement, for benches and garden borders.

🔬 The science Adults level

Eco-bricks are plastic sequestration: they hold plastic out of the environment, but do not recycle it or reduce production. Treat them as the last step after refuse, reduce and reuse. The Global Ecobrick Alliance sets a minimum density of 0.33 g/mL for building use. The woven-wrapper bag (from a 2024 student proposal) upcycles multilayer packaging that has almost no recycling market.

> 👩‍🏫 **Teacher tip:** Agree on ONE bottle size for the whole school so the bricks fit together. A 1.5 L brick takes several sessions: let pupils take bottles home. Keep a class log of the total plastic packed (kg).

🚀 Challenge

1. Class total: add up the mass of all the eco-bricks. How many plastic bags is that? (Weigh 10 bags to find the mass of one.)
2. Design: sketch a bench made of 1.5 L eco-bricks. How many bricks do you need? How much plastic would it lock away?
3. Check a real design: a 2024 student stool used 21 bottles of 1.5 L and 800 g of plastic in total. What was the density of each brick? Does it meet the 0.33 g/mL minimum? How much plastic would 21 proper bricks hold?
4. Cost it like an engineer: A 2024 student team spent RM 10.50 on glue sticks and tape for each stool, plus a measuring tape (RM 12) and a glue gun (RM 8) bought once. They planned to sell at RM 25. What is the profit on the first stool? On the tenth?
5. Bag test: load your woven bag with books until it fails. Record the mass. Where did it break, and how could the design be stronger?

💭 Think about it

- Which plastics in your home are hardest to recycle?
- Is an eco-brick recycling? Why or why not?
- How could your school use less soft plastic in the first place?



Eco-brick stool and pillar made by students. Photos and posters: UPM chemical engineering students, Faculty of Engineering.

✅ Quick quiz

1. What is an eco-brick?

A. A brick made by melting glass bottles in a kiln

B. A bottle packed hard with clean, dry plastic, used for building

C. A block of dried compost pressed into a mould

2. Why must plastic stuffed into an eco-brick be clean and dry?

A. Wet plastic packs in tighter and stronger

B. Clean plastic makes the bottle lighter

C. Leftover food would rot, smell and grow germs

3. Resin code 1 (♳) on a drink bottle stands for…

A. PVC (polyvinyl chloride)

B. PS (polystyrene)

C. PET (polyethylene terephthalate)

D. PP (polypropylene)

Answers: 1 B · 2 C · 3 C



Lab 9

Composting: Bokashi & Worms

Feed food scraps to microbes and worms and get rich soil back.

Ages 7+ | ⏱ 48 min to set up; 2-8 weeks to finish | Difficulty ★★☆

Light supervision | 🌿 No heat | SDG 2 | SDG 12 | SDG 13 | SDG 18

Waste used
Food scraps, dry leaves, cardboard

Product
Compost, vermicast and liquid plant feed

Cost
Low: reuse old buckets. Example shop prices (Oct 2026): bokashi bran about RM 8–14 per kg. Composting worms are sold by the kilogram and cost much more (one seller: RM 380 per kg in Peninsular Malaysia); the price of a small starter amount varies.



Scan to watch

🌏 Why it matters

🌍 Environment

Food is 30.6% of Malaysian household waste. Buried in landfill without air, it rots into methane (a strong greenhouse gas) and dirty liquid called leachate. Composting turns it back into soil instead.

💰 Economy

Compost and worm castings replace some bought fertiliser and potting soil, and gardeners will buy them.

🤝 Society

A school compost corner links the canteen, the garden and the classroom; vegetables grown with it can go back to the canteen.

🧺 You need

Bokashi bin

- A bucket with a tight lid, ideally with a tap at the bottom (or two stacked buckets, the inner one with drainage holes drilled by an adult)
- Bokashi bran: about a handful per layer
- Food scraps cut small: fruit and vegetable scraps, cooked rice and noodles, a little cooked meat or fish
- A plate or a bag of sand to press the scraps down
- A garden bed or compost heap for the finished bokashi

Worm bin

- A plastic box with a lid, raised on two bricks over a tray
- Bedding: shredded newspaper or cardboard (not glossy), dry leaves, a handful of garden soil
- Composting worms suited to the tropics, such as the African nightcrawler (Eudrilus eugeniae), from a vermicompost farm. Garden earthworms do not live well in bins.
- Fruit and vegetable scraps, crushed eggshells, used coffee grounds
- A spray bottle of water and gloves

🪜 Steps

kashi bin

1. Cut the food scraps into thumb-sized pieces and drain off any soup or liquid.
2. Sprinkle a handful of bokashi bran over the bottom of the bucket.
3. Add a layer of scraps, sprinkle bran on top, then press down hard to squeeze out the air.

   🔬 Bokashi microbes work without air; air pockets let mould and bad smells grow.

4 Close the lid tightly every time. Repeat the layers until the bucket is full.

5 Every 2–3 days, drain the liquid from the tap. Use it only diluted, at least 1:100 (10 mL in 1 litre of water), to water plants.

6 When full, keep it sealed for 2 weeks. It should smell sour, like pickles. White fluffy mould is fine; black or green mould or a rotten smell means air or too much water got in.

7 Bury it in a trench about 20 cm deep, or mix it into a compost heap. After 2–4 weeks it has turned into soil.

🔬 Bokashi is a fermented "pre-compost": it finishes breaking down in the soil.

rm bin

8 ADULT: drill small air holes in the lid and sides and drainage holes in the bottom. Stand the box on bricks over a tray.

9 Fill half the box with damp bedding plus a handful of soil. Squeeze it: it should feel like a wrung-out sponge.

10 Gently put the worms on top and leave them a day to settle in. Keep the bin in the shade, never in direct sun or rain.

11 Feed a small amount of chopped scraps at a time, buried under the bedding in a different corner each time. Cover with a handful of dry leaves or paper.

🔬 Aim for about 2–3 parts dry "browns" (leaves, paper) to 1 part wet "greens" (food scraps) by volume.

12 Do not feed worms meat, fish, dairy, oily or salty food, or much citrus, onion, garlic or chilli.

13 After 1–2 months the bottom is dark, crumbly vermicast. Push everything to one side and put fresh bedding and food on the other. In 1–2 weeks the worms move across; scoop out the castings.

14 Mix a little vermicast into potting soil or sprinkle it around your plants.

🦺 Safety First

⚠️ Wear gloves or wash your hands well after touching compost, worms or bokashi. Cover any cuts. Open bags of compost, potting mix or bokashi bran away from your face, and do not breathe in the dust.

⚠️ Mould: do not put your face close to a mouldy bin or sniff it. People with asthma or allergies should let someone else open the bin.

⚠️ Bokashi liquid is sour and strong: always dilute it, never drink it, and keep lids shut so flies stay away.

⚠️ Empty the worm-bin tray regularly: standing water breeds Aedes mosquitoes. An adult drills the holes.

⚠️ No pet poo, diseased plants or "compostable" plastic bags: most need hot industrial composting.

🔬 The science Adults level

In landfill, food waste decomposes anaerobically and produces methane, a far more potent greenhouse gas than CO_2, plus leachate. Separating food waste at source and treating it at home, school or community level cuts collection costs and landfill load. Bokashi takes cooked food and small amounts of meat (which worm bins and open heaps should not), needs little space and suits flats; worm farms produce high-value vermicast. Note: the 2025 poster's "1:3:2 NPK" was replaced with a browns-to-greens guide, because NPK describes

a fertiliser's nutrient content, not a mixing ratio.

> 👩‍🏫 **Teacher tip:** Set up both systems early in the term with a care rota. Buy worms from a local vermicompost farm. Compare after a month: which system handled the canteen waste better?

🚀 Challenge

1. Weigh the food scraps your class adds every day for 2 weeks. How many kg would you keep out of landfill in a school year?
2. Fair test: grow two pots of kangkung, one with vermicast mixed in and one without. Measure the height every week.
3. Compare bokashi and worms: cost to set up, space, smell, speed and what each can accept. Which suits your home?

💭 Think about it

- Which foods can go in bokashi but not in a worm bin?
- Why does food buried in landfill make methane?
- How did it feel to look after living worms?

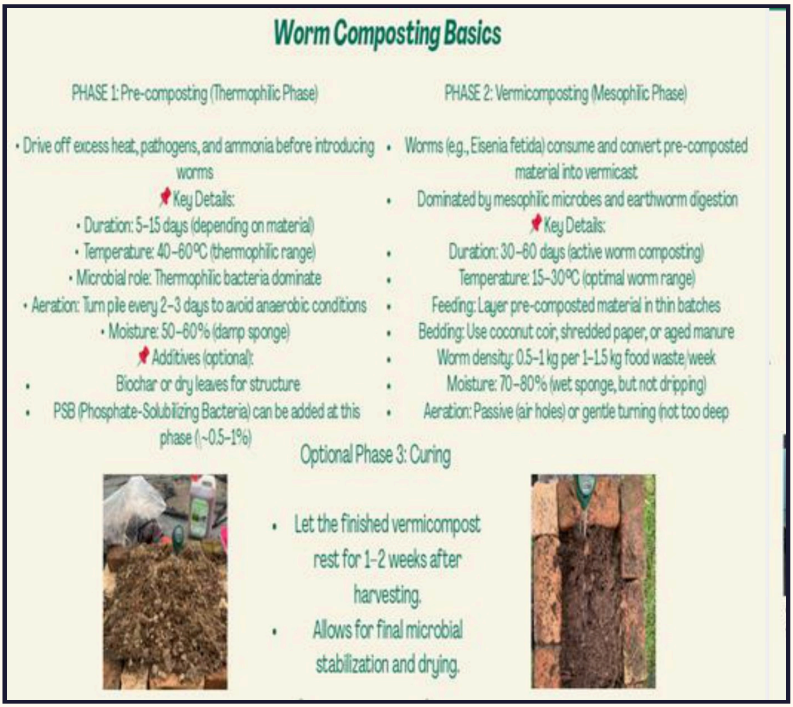
✅ Quick quiz

1. What is bokashi?

A. Burning dried food waste in a small stove for energy

B. Fermenting food waste in a sealed bucket with microbe bran

C. Composting food waste in an open pile using only worms



Worm composting poster. Photos and posters: UPM chemical engineering students, Faculty of Engineering.

2. Which animal helps make vermicompost?

A. Earthworms

B. Ants

C. Cats

3. Vermicomposting usually uses...

A. Garden snails and slugs

B. Composting worms (red wigglers)

C. Soil ants and termites

Answers: 1 B · 2 A · 3 B



Lab 10

🌽 Cornstarch Bioplastic

Cook starch, water, vinegar and glycerin into a plastic you can shape, then test whether it really is "green".

Ages 9+ | ⏱ 60 min + 2 days drying | Difficulty ★★☆

Adult does hot or sharp steps | 🔥 Needs heat | SDG 9 | SDG 12 | SDG 14

🧒 **Age and supervision:** An adult does the heating on the stove. Children measure, stir away from the heat and shape the cooled mixture.

Waste used

Not a waste product: a plant-based swap for single-use plastic

Product

Seedling pots, coasters and small shapes

Cost

About RM 0.84 per small pot (2024 UPM student estimate; prices may have changed).



Scan to watch

🌍 Why it matters

🌍 Environment

Plastic is 21.9% of Malaysian household waste, and most plastic is made from oil. Starch plastic comes from plants, but "made from plants" does not automatically mean it breaks down anywhere. This lab tests that honestly.

💰 **Economy**

Starch, glycerin and vinegar are cheap. A 2024 UPM student team costed a seedling pot at about RM 0.84.

🤝 **Society**

Students act as materials engineers: they design, test and judge a "green" claim with evidence.

🧺 You need

- 15 g cornstarch (tepung jagung), about 2 tablespoons
- 100 mL water
- 10 mL (2 teaspoons) white vinegar
- 5–10 mL (1–2 teaspoons) glycerin, from a pharmacy: more glycerin = bendier plastic
- 1–2 drops food colouring (optional)
- Small saucepan, heat-proof spatula, measuring spoons and jug
- Baking paper on a tray, or a small cup or silicone mould to shape a pot
- Stove (adult) and labels

🪜 Steps

1. Measure all the ingredients before any heat is turned on.
2. Put the water, cornstarch, vinegar, glycerin and colour into the cold saucepan. Stir until there are no lumps.

3. ADULT: heat on low, stirring all the time.
4. Watch it change: milky white, then thick, clear and gel-like (about 5–15 minutes). Turn off the heat.

   🔬 This change is gelatinisation: the starch grains swell, burst and release their long chains.
5. ADULT: spread the hot gel on baking paper about 3–5 mm thick, or press it around the outside of a small cup to make a pot. It is very hot and sticky.
6. When it has cooled to warm (not hot: check with the back of a spoon first), pop any air bubbles with a spoon and smooth the top.
7. Leave it to dry for at least 2 days in an airy place. Turn sheets over after the first day.
8. Peel it off and test it: bend it, stretch it, and put a small piece in water for an hour. What happens?

   🔬 Label each sample with its glycerin amount so you can compare.

🦺 Safety First

⛔ Hot starch gel sticks to skin and burns like hot porridge. Only an adult heats, spreads and moulds it; children mix the cold ingredients and do the testing.

⚠️ For a burn: cool it under cool running water for 20 minutes and tell an adult.

⚠️ Not food, even though the ingredients are: label it and do not eat it.

⚠️ Do not put it in the plastic recycling bin: it contaminates real plastic recycling. Small pieces can go in the compost.

🔬 The science Adults level

"Bioplastic" mixes two separate ideas. Bio-based means made from plants; biodegradable means microbes break it down under stated conditions. They are independent: bio-based polyethylene is not biodegradable, while some fossil-based polyesters are. ASTM D6866 measures bio-based carbon content (by radiocarbon); it says nothing about biodegradability, which is certified under standards such as EN 13432 or ASTM D6400. Plain starch films are weak and water-sensitive, so they suit short-life uses such as seedling pots, not packaging for wet food. The 2024 student costing of RM 0.84 per pot is a good start; scaling would need consistent drying, moisture resistance and a real end-of-life route.

👩‍🏫 **Teacher tip:** Pre-measure the dry ingredients into cups. One saucepan per adult. A microwave also works: heat in 20-second bursts, stirring between (adult only).

🚀 Challenge

1. Glycerin test: make films with 5, 10 and 15 mL glycerin. How far does each bend before it cracks? Which would make the best seedling pot?
2. Breakdown test: bury one piece in soil, keep one in a jar of water and one dry. Photograph them every week for 4 weeks.
3. Cost it: price one pot (starch + glycerin + vinegar) and compare with a plastic seedling pot. Would a plant nursery switch?

💭 Think about it

- Is "made from plants" the same as "breaks down in nature"?
- What would you use your bioplastic for, and what should it never be used for?
- How could you make it stronger or more water-resistant?



Bio-pots made by students. Photos and posters: UPM chemical engineering students, Faculty of Engineering.

✅ Quick quiz

1. Which ingredients make a simple cornstarch bioplastic?

A. Cornstarch, water, vinegar and glycerol

B. Cornstarch, sand, cement and salt

C. Flour, petrol, salt and sugar

2. In cornstarch bioplastic, what does glycerol do?

A. It makes the film hard and brittle

B. It is a plasticiser that makes the film flexible

C. It is a dye that gives the film colour

3. Which resin code is HDPE (e.g. shampoo and detergent bottles)?

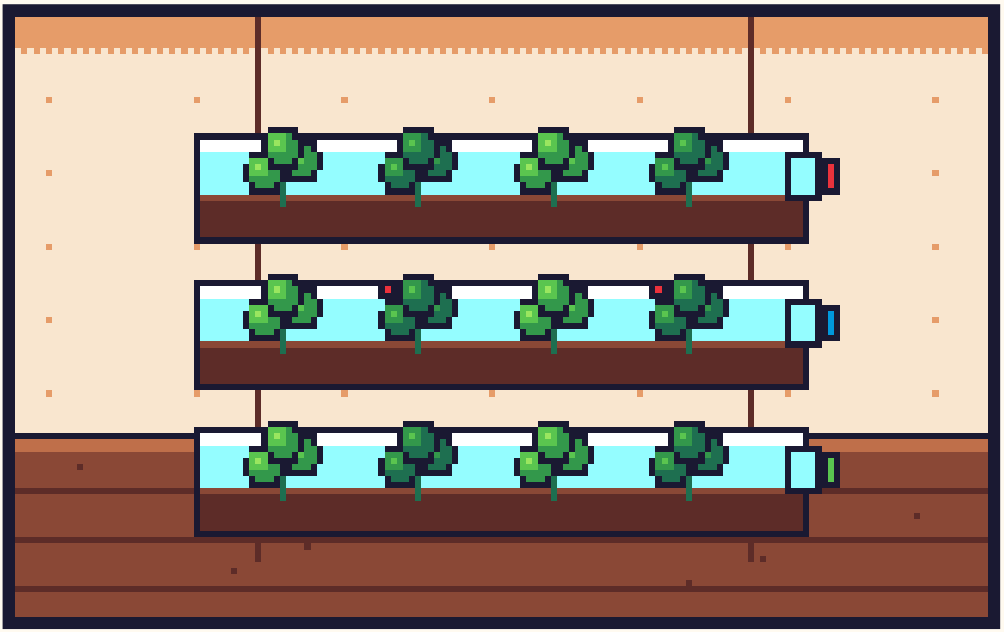
A. 4

B. 6

C. 7

D. 2

Answers: 1 A · 2 B · 3 D



Lab 11

Bottle Vertical Garden

Hang a garden of plastic bottles on a wall or fence: water the top one and it drips down to the rest.

Ages 7+ | ⏱ 60 min to build, then water a little every day | Difficulty ★☆☆ | Close supervision | 🌿 No heat | SDG 2 | SDG 11 | SDG 12 | SDG 13 | SDG 15

Waste used
1.5 L PET bottles; compost from food waste

Product
A hanging vertical garden and mini bottle pots

Cost
Almost free: used bottles and string. If you buy soil and seeds, example shop prices (Oct 2026): a 7 L bag of potting mix RM 7.50, a vegetable seed packet RM 5.90. Prices vary.



Scan to watch

🌏 Why it matters

🌍 Environment

Each bottle gets a second life before recycling, and plants on walls cool hot surfaces and give insects food.

💰 Economy

Grow kangkung, herbs or cuttings to use or sell, in a space too small for a normal garden.

🤝 Society

A green wall makes a school corner or flat balcony nicer, and mini pots make easy gifts.

🧺 You need

- 4–6 clear 1.5 L bottles with caps, washed, labels off
- Strong string or nylon cord, and a frame, fence or grille to hang from
- Soil mixed with compost (see the Compost lab)

- Plants: leafy seedlings (e.g. kangkung), herb cuttings, or money plant cuttings
- Craft knife or strong scissors (adult only), a skewer, tape and a marker

🪜 Steps

1. Lay a bottle on its side. Draw a window about 15 cm × 6 cm on the top side.
2. ADULT: cut out the window and tape the cut edges so they are not sharp.
3. ADULT: poke 3–4 small drainage holes on the bottom side with a skewer.

   🔬 Without holes, water collects at the bottom and the roots rot because they cannot get air.
4. Tie string around both ends (near the neck and the base) to make a hanger. Do the same for every bottle.
5. Hang the bottles one below another, about 25 cm apart, so water dripping from one falls into the next.
6. Fill each bottle two-thirds full with the soil and compost mix.
7. Plant your seedlings or cuttings and press the soil gently around them.
8. Water the top bottle slowly and watch it trickle down. Place the garden where it gets a few hours of sun.
9. Mini pots: cut the bottom 8 cm off spare bottles, poke holes, fill and plant. Tie a ribbon on for a gift.

   🔬 The student team also built an Arduino soil sensor and pump to water the garden automatically: see the Smart Watering lab.

🦺 Safety First

⚠️ Cut plastic edges are sharp: an adult cuts, then tape the edges.

⚠️ Wet soil is heavy. Tie the bottles to a strong frame, not above where people walk or sit. An adult hangs anything above head height: never climb on chairs or railings.

⚠️ Water drips down the column: hang the garden away from plugs, sockets and extension cords.

⚠️ No standing water: Aedes mosquitoes can breed in a cap or tray in about a week. Check and empty them weekly.

⚠️ Money plant is for looking at only: it is poisonous to eat and harmful to cats and dogs. Wash hands after gardening.

🔬 The science Adults level

Green walls reduce surface temperatures and support urban biodiversity, and reusing PET keeps it in use longer before recycling. Sunlight slowly breaks down PET, so expect to replace bottles after some time outdoors, and recycle the old ones.

🧑‍🏫 **Teacher tip:** Prepare the cut bottles before class. Pupils fill, plant and hang. Make a watering rota and a weekly mosquito check. Pair it with the Compost lab for the soil and the Smart Watering lab for older pupils.

🚀 Challenge

1. Measure: how many mL of water do you pour at the top, and how many mL drip out of the bottom bottle? Where did the rest go?
2. Grow the same seedling in a bottle and in the ground. Measure the height every 3 days for 3 weeks and draw a graph.
3. Count how many bottles a class garden of 20 bottles saves from the bin in a year if you replace them once.

💭 Think about it

- Where at home or school could a bottle garden go?
- Is reusing a bottle better than recycling it? Why?
- What would you grow, and who would you give it to?

✅ Quick quiz

1. How does a self-watering plant pot made from a plastic bottle keep the soil moist?

A. A cloth wick pulls water up from below

B. The plastic slowly turns into water

C. Sunlight pumps water up through the cap



Seedlings in mini bottle pots. Photos and posters: UPM chemical engineering students, Faculty of Engineering.

2. In a bottle vertical garden, why do the bottles need small holes underneath?

A. So insects can get in to eat the plants

B. So the bottle is lighter to carry

C. So extra water drains and the roots get air

3. A school compost garden that makes the soil healthy supports which SDG?

A. SDG 1 No poverty

B. SDG 15 Life on land

C. SDG 16 Peace and justice

Answers: 1 A · 2 C · 3 B



Lab 12

Bottle Hydroponics

Grow leafy vegetables in water, with no soil and no pump, inside a cut plastic bottle.

Ages 9+ | ⏱ 60 min to set up; about 4-6 weeks to harvest | Difficulty ★★☆

Close supervision | 🌿 No heat | SDG 2 | SDG 6 | SDG 11 | SDG 12

Waste used
1.5 L PET bottles and caps; scrap wood and bubble wrap for the shelter

Product
A soil-free vegetable pot and a mini rain shelter

Cost
Bottles are free; hydroponic nutrient (AB mix) and seeds are bought. Example (Oct 2026): a vegetable seed packet RM 5.90; AB mix price varies by pack size (one listing showed RM 16.90).



Scan to watch

🌏 Why it matters

🌍 Environment

Bottles get reused, and a hydroponic pot with no pump uses no electricity. Plants take only the water they need.

💰 Economy

Fresh sawi or lettuce at home or in the school canteen, grown on a wall or a corridor with no garden land.

🤝 Society

Flats and schools without land can still grow food. A 2024 student team designed a "rain shelter house" so the bottles keep working in the rainy season.

🧺 You need

Bottle pot

- 1 clear 1.5 L bottle per plant, washed

- Seeds of a leafy vegetable (sawi, lettuce or kangkung) and a kitchen sponge cut into 2 cm cubes
- Hydroponic nutrient A and B (AB mix) and a measuring syringe or spoon
- Black paper, foil or old cloth to cover the bottle; tape
- Craft knife (adult only) and a marker

Rain shelter (optional)

- Scrap wood strips, nails and hammer; extra bottles; stapler; clear tape; bubble wrap

🪜 Steps

ttle pot (Kratky method)

1. Sow 1–2 seeds on each wet sponge cube. Keep them damp in a tray until roots and 2 small leaves appear (about a week).
2. ADULT: cut the bottle around one-third from the top. Tape the cut edges.
3. Wrap the bottom part in black paper or foil, leaving a thin window to check the water level.

   🔬 Light makes green algae grow in the water. Algae steal nutrients and oxygen from the roots.
4. ADULT: mix the AB nutrient into water exactly as the label says. Add A and B to the water separately, never to each other.
5. Pour the nutrient water into the bottom part. Turn the top part upside down (no cap) and sit it in the bottom like a funnel.
6. Push the sponge with the seedling into the bottle neck so its base just touches the water.
7. Tape around the joint so no light, rain or mosquitoes can get in. Label the bottle "Not for drinking".

8 Keep it in bright light but out of the rain. Do not refill to the top: as the roots grow, the water level drops and leaves an air gap.

🔬 The top roots breathe from the air gap and the lower roots drink. If the water gets very low, top up only to about half.

9 Harvest the leaves in about 4–6 weeks. Pour leftover nutrient water onto garden plants, not into a drain.

in shelter house

10 Build a small wooden frame to fit the bottles you collected. Nail each bottle cap to the wood, then screw the bottle into its cap so it stays steady.

11 For the roof, ADULT cuts the top and bottom off bottles and slits them open. Flatten, staple them together and tape the joins so rain cannot leak through.

🔬 The student team used bubble wrap to fill gaps and save cost, and fixed the roof without glue.

🦺 Safety First

⚠️ Nutrient salts: an adult measures them. Do not drink or taste, wash hands after, and keep the packets away from small children.

⚠️ Still water can breed Aedes mosquitoes in about a week. Seal every gap with tape and check for wrigglers weekly. If you see any, empty the bottle and start again.

⚠️ Cut bottle edges, nails and hammers: adults cut and hammer, learners wear covered shoes.

🔬 The science Adults level

Non-circulating (Kratky) hydroponics was developed at the University

of Hawaii for low-cost growing without electricity. It suits schools and flats, but nutrient solutions are fertiliser: spent solution should go on plants, not into drains, where it feeds algae blooms.

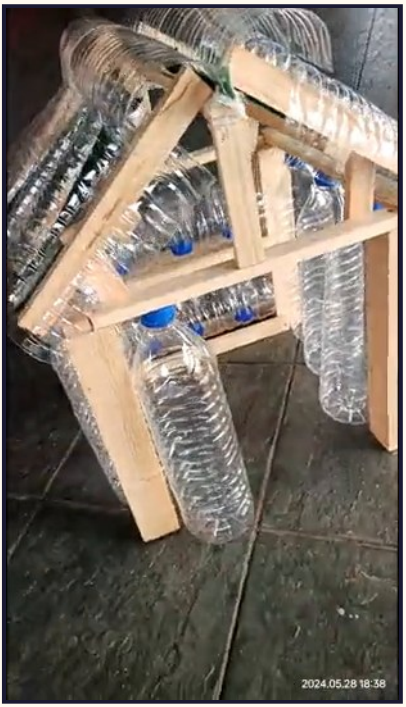
> 👩‍🏫 **Teacher tip:** Sow the sponges a week before the lesson so pupils start with seedlings. One bottle per pair is enough. Keep a class table of the water level and leaf count each week.

🚀 Challenge

1. Grow two bottles: one wrapped in black paper, one clear. After two weeks, which has more algae? Which plant is bigger?
2. Mark the water level every 2 days. How many mL does one plant drink a day?
3. Design a rain shelter for 10 bottles on your school wall. Sketch it with measurements and list the waste materials you would use.

💭 Think about it

- Why might hydroponics help people who live in flats?
- What would happen to the roots if the bottle was filled to the top all the time?
- Why did the student team need a roof over their bottles?



A rain shelter house prototype. Photos and posters: UPM chemical engineering students, Faculty of Engineering.

✅ Quick quiz

1. In the Kratky bottle hydroponics method, why should you NOT keep refilling the bottle to the top?

A. The upper roots need an air gap to get oxygen

B. Plants can only drink from a half-empty bottle

C. Nutrients only dissolve in a small amount of water

2. Which statement about “bio-based” plastics is correct?

A. All bio-based plastics dissolve in the sea within days

B. Bio-based means made from plants; it is not always biodegradable

C. Bio-based plastics are always made from petroleum

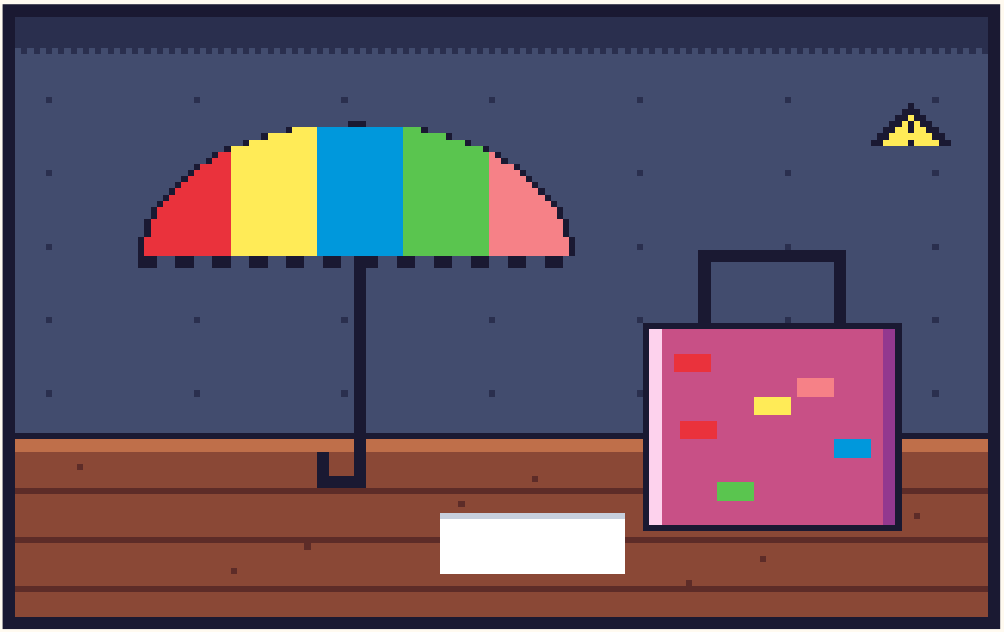
3. Why can “compostable” plastic items cause problems in normal plastic recycling?

A. They make recycled plastic too strong to mould

B. They contaminate the stream and weaken recycled plastic

C. They turn the recycling plant into a compost site

Answers: 1 A · 2 B · 3 B



Lab 13

☂️ Fused-Plastic Bags & Mini Umbrella

Iron used plastic bags (between baking paper) into a tough new fabric, then make a drawstring bag, a tote or a working mini umbrella.

Ages 9+ | ⏱ 90 min (the umbrella takes a second session) | Difficulty ★★☆

Adult does hot or sharp steps | 🔥 Needs heat | SDG 12 | SDG 14

🧒 **Age and supervision:** Primary pupils (up to 12): an adult does all the ironing as a demonstration, or brings ready-made fused sheets. Children cut, design and sew.

Waste used
Thin plastic carrier bags (marked 2 HDPE or 4 LDPE); plastic straws

Product
Fused plastic fabric: drawstring bag, tote bag, mini umbrella

Cost
Almost free: used bags and straws, plus thread and string. Baking paper: example rolls from about RM 5.30 (30 cm × 5 m) to RM 9.90 (38 cm × 12 m); prices vary.



Scan to watch

🌏 Why it matters

🌍 Environment
Carrier bags are light, blow into drains and are rarely recycled. Fusing them turns many weak bags into one strong,

💰 Economy
Waterproof pouches and totes from free waste can be sold at school fairs.

🤝 Society
Sewing and designing build real skills, and each bag carries a message about plastic.

🧺 You need

- 5–10 used carrier bags, clean and dry (check the symbol: 2 or 4 only)
- Baking (parchment) paper, a clothes iron and an ironing board or towel-covered table
- Scissors, ruler and marker
- Needle and thread or a sewing machine; buttons
- For the umbrella: about 10 plastic straws

🪜 Steps

.ke the fabric (ADULT irons)

1. Cut off the handles and the bottom seam of each bag, then cut down one side so it opens into a flat sheet.
2. Stack 4–6 layers of bag on baking paper and cover with another sheet of baking paper. Plastic must never touch the iron.
3. ADULT: iron on a low-medium setting with no steam, keeping the iron moving for 15–20 seconds. Turn over and repeat.

   ⚠️ **Too cool and the layers will not join (as one student team found); too hot and it shrinks, gets holes and smells. Stop if it smokes.**
4. Let it cool, then peel off the paper. The bags have shrunk into one stiff, waterproof sheet.

awstring bag

5. Measure, mark and trim the fabric to two equal rectangles.
6. Fold 2 cm over at the top edge, cover with baking paper and ADULT irons the fold to seal a tunnel. Then seal the sides and bottom the same way, or sew them.

7 Make a plastic-bag cord: fold a bag 3 times, cut strips two fingers wide, open them out and cut in a zig-zag to get one long strip. Thread it through the tunnel and knot.

te bag

8 Cut fabric pieces to the size you want (front, back, base, two handles). Sew them together by hand or machine and add a button.

ni umbrella

9 ADULT irons straws flat between baking paper to make stiff strips. Cut each strip in half for 2 ribs, or in quarters for 4 stretchers.

🔬 Most straws are polypropylene (code 5), which softens at a higher temperature than bag plastic. Raise the heat a little at a time, keep the window open and stop if it smokes.

10 Bundle 4 straws as the shaft. Make 2 small holes at the ends of each rib and stretcher.

11 Sew the stretchers to a short straw sleeve (the runner) with needle and thread, and the ribs to the top of the shaft. Add a stopper so the runner locks open.

12 Open and close it to test, then cut a circle of fused fabric as the canopy and stitch it to the rib tips.

🦺 Safety First

⛔ ADULT ONLY for all ironing. Open windows or use a fan, keep the iron on low-medium, always use baking paper on both sides, and stop at once if there is smoke or a sharp smell.

⛔ Use only bags marked 2 (HDPE) or 4 (LDPE). Never heat PVC (3), foil-lined wrappers or bags with metal print: they can give off toxic fumes.

⚠️ Hot plastic sticks to skin and burns. Let it cool before touching. Stand the hot iron upright, out of reach, and switch it off when you finish.

⚠️ Needles and scissors: count needles before and after, and keep fingers clear of the sewing-machine needle.

🔬 The science Adults level

Fusing is upcycling by heat, so the controls matter: thin PE melts at roughly 115–135 °C, well below the temperature at which it decomposes, but PVC releases hydrogen chloride when heated and printed or foil layers can release other fumes. Ventilate, keep temperatures low and sort by resin code first. Once fused and sewn, the product can still go back into PE recycling at end of life if no other materials are attached.

🧑‍🏫 **Teacher tip:** The adult runs one ironing station by an open window; groups queue with their stacked bags and baking paper. Others cut, design and sew. Make the fabric in session 1 and the umbrella in session 2.

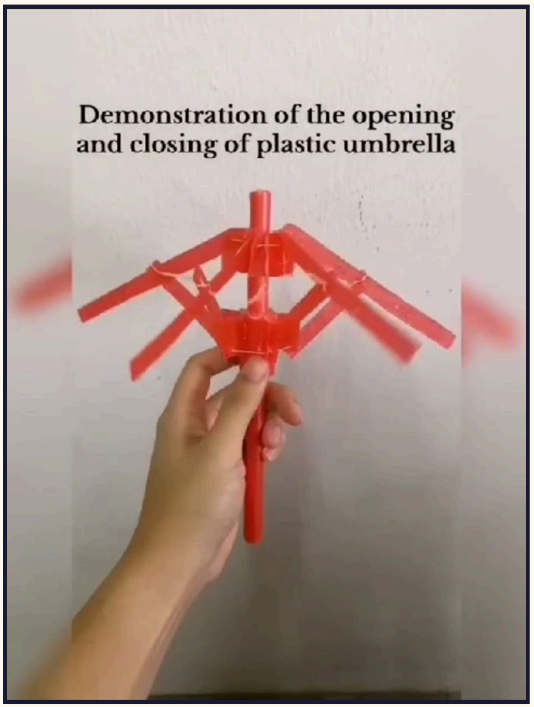
🚀 Challenge

1. Strength test: hang a bag of 1 layer, 4 layers and 8 layers from a hook and add water bottles until each tears. Plot layers vs. load.
2. Waterproof test: pour 100 mL of water into your drawstring bag. Does any leak out in 5 minutes? Where?

3. Count how many carrier bags went into your product. How many would your whole class save in a term?

💭 Think about it

- Why must plastic never touch the iron directly?
- Is a fused-plastic bag better than refusing the carrier bag in the first place?
- What else could you make from the fabric?



Mini umbrella frame made from ironed straws. Photos and posters: UPM chemical engineering students, Faculty of Engineering.

✅ Quick quiz

1. Resin code 4 is LDPE. Which product is typically LDPE?

A. Polystyrene food boxes

B. Plastic bags and cling film

C. PVC water pipes

2. Which plastic must NEVER be ironed or heated to make fused plastic fabric?

A. LDPE carrier bags (code 4)

B. PVC (resin code 3)

C. HDPE carrier bags (code 2)

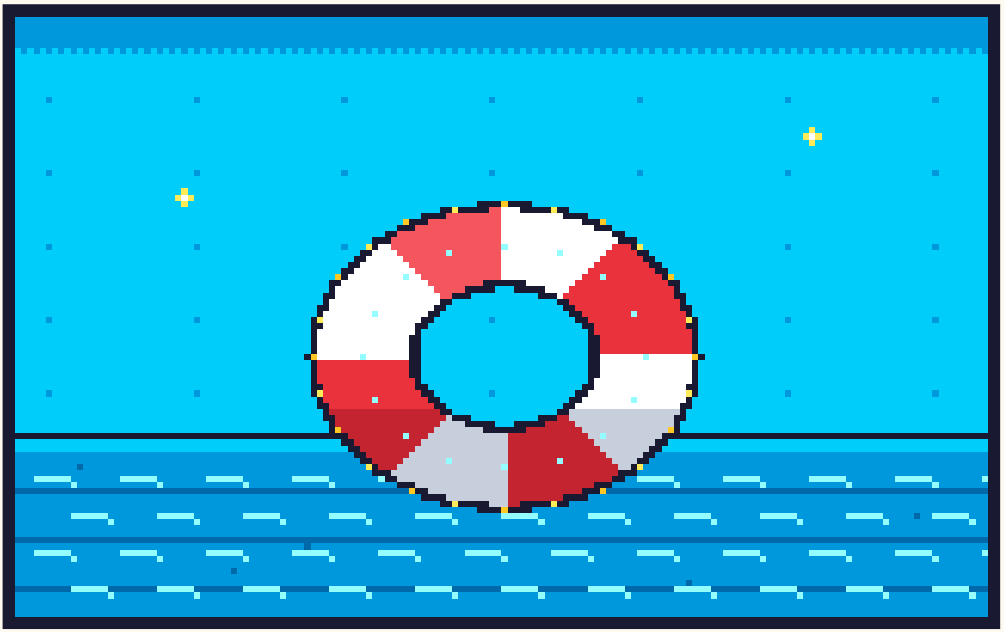
3. How long can a plastic bottle last in nature?

A. About one week, then it melts away

B. Hundreds of years, breaking into tiny bits

C. A few months, like a banana peel

Answers: 1 B · 2 B · 3 B



Lab 14

🛟 Bubble-Wrap Float Ring ⟨Model⟩

Roll parcel bubble wrap into a ring that floats, then test how much weight it can hold up in a basin.

Ages 7+ | ⏱ 45 min | Difficulty ★☆☆ | Close supervision | 🌿 No heat

SDG 11 | SDG 13 | SDG 14

> 🧒 **Age and supervision:** This is a model for learning how things float. It is not a safety device: never use it in water to hold up a person.

Waste used
Bubble wrap from parcels; old PE tarpaulin or plastic bags

Product
A model float ring for a buoyancy experiment

Cost
Free: parcel bubble wrap, a scrap of tarpaulin or a plastic bag, and tape.



Scan to watch

🌏 Why it matters

🌍 Environment

Online shopping produces lots of bubble wrap and plastic film. Kept in use, it stays out of drains, where plastic blocks water flow and makes flash floods worse.

💰 Economy

Free materials make a full science experiment on floating and sinking.

🤝 Society

Floods are Malaysia's most common natural disaster. Building a model opens a talk about flood safety: what to do, and what never to do.

🧺 You need

- A long piece of bubble wrap (about 1 m × 30 cm for a model)
- Wide tape (packing or duct tape)

- An old PE tarpaulin scrap or 2 plastic bags for the cover
- A basin or big tub of water, and weights: 500 mL water bottles, coins or marbles
- Kitchen scale and ruler

Students' full-size design	Value
For a child up to	30 kg
Inner radius	0.25 m
Outer radius	0.57 m
Tube radius	0.16 m
Volume	≈ 0.2 m³

From the 2024 lifebuoy team's slides. A model, not a tested safety device.

🪜 Steps

1. Lay 3 layers of bubble wrap on top of each other and tape them together into one thick sheet.
2. Roll the sheet tightly into a long tube and tape along its length.
3. Bend the tube into a ring and tape the two ends firmly together.
4. Cover the ring with strips of tarpaulin or plastic bag and tape them down. This is your waterproof skin.
5. Weigh the ring. Measure its outer and inner diameter and the thickness of the tube.
6. Float it in the basin. Add weights one at a time until water reaches the top of the ring. Record the total mass it held.

   🔬 1 litre of water has a mass of 1 kg. A ring that can push aside 2 litres of water can hold up about 2 kg, minus its own mass.

🦺 Safety First

⛔ This is a learning model, NOT a life-saving device. Never use it for swimming, in a flood or for a rescue. Only certified lifejackets and lifebuoys are safe.

⛔ In a flood: stay out of the water, move to higher ground and call 999. If someone is in the water, REACH or THROW, don't GO in.

⚠️ An adult stays by the basin the whole time: young children can drown in a few centimetres of water. Empty it after use.

⚠️ Keep bubble wrap and plastic bags away from babies and toddlers (suffocation).

🔬 The science Adults level

Volume is not the only test of a life-saving device. A certified ship's lifebuoy (IMO Life-Saving Appliances Code) must hold up at least 14.5 kg of iron in fresh water for 24 hours, survive a drop into water and resist fire, oil and sunlight. Bubbles can burst and seams can leak, which is why DIY floats stay as models. The real lesson is prevention: plastic kept out of drains reduces flood risk.

👩‍🏫 **Teacher tip:** Run it as a fair test: groups change one thing (layers, tube thickness or ring size) and compare the mass held. Finish with the flood safety rules and "Reach or throw, don't go".

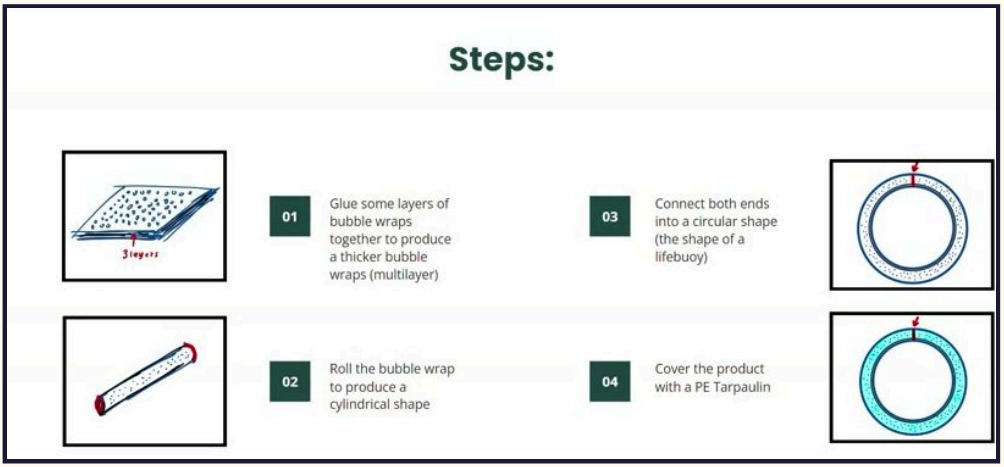
🚀 Challenge

1. Use $V = 2\pi^2Rr^2$ to predict how much your model should hold. Compare with your test. Why are they different?

2. Check the students' numbers: with R = 0.41 m and r = 0.16 m, what volume do you get?
3. Make a poster for your neighbourhood: "Keep plastic out of drains" and the flood safety rules.

💭 Think about it

- Why is a home-made float not safe for a real flood?
- How does plastic in drains make floods worse?
- What would you do if you saw someone in floodwater?



4 steps: layer, roll, join, cover. Photos and posters: UPM chemical engineering students, Faculty of Engineering.

✅ Quick quiz

1. A bubble-wrap float ring holds up 2 kg in a basin. What is it safe to use it for?

- A. Rescuing a child in a flood
- B. Swimming in the sea
- C. Only as a science model, never for swimming or floods

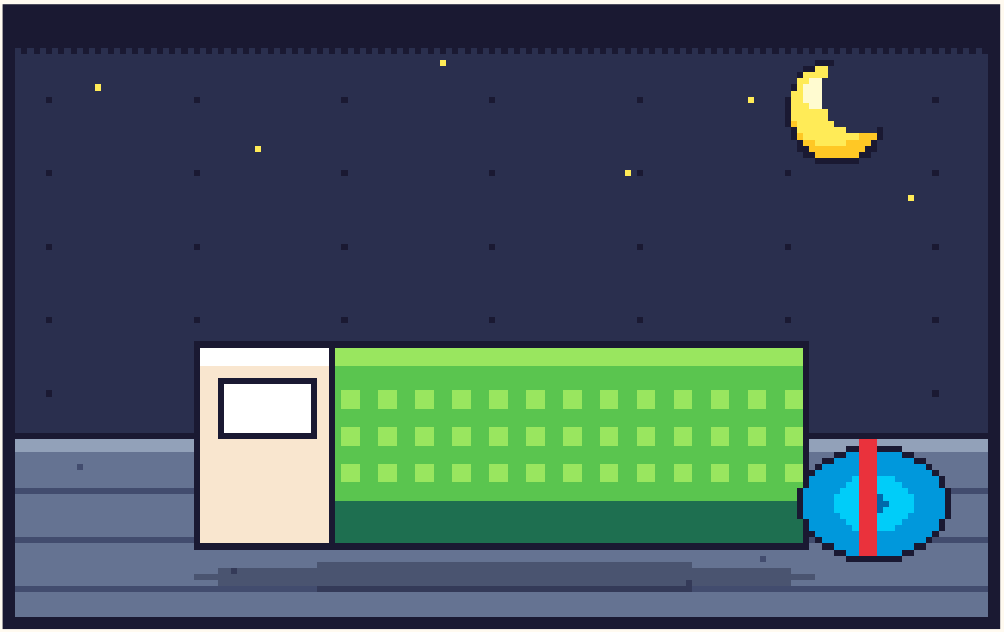
2. Sea turtles sometimes eat plastic bags because…

- A. Floating bags look like jellyfish, their food
- B. Plastic tastes sweet
- C. Turtles collect plastic for nests

3. Why does a bubble-wrap mat keep you warmer on a cold floor?

- A. The trapped air in the bubbles is bad at carrying heat
- B. Plastic makes its own heat
- C. The bubbles reflect all the cold away

Answers: 1 C · 2 A · 3 A



Lab 18

🛌 3-in-1 Bubble-Wrap Sleeping Bag

Sew squares of bubble wrap and old cloth into a mat that becomes a blanket or a buttoned sleeping bag.

Ages 10+ | ⏱ About 2 hours, over 2-3 sessions | Difficulty ★★☆

Close supervision | 🌿 No heat | SDG 1 | SDG 3 | SDG 11 | SDG 12

🧒 **Age and supervision:** A learning prototype only: it has not been tested for outdoor or emergency use. An adult supervises needles and any sewing machine.

Waste used
Parcel bubble wrap; old bedsheets, curtains or T-shirts

Product
A 3-in-1 mat, blanket and sleeping bag

Cost
Almost free: bubble wrap from parcels and old cloth; thread and buttons.



Scan to watch

🌏 Why it matters

🌍 Environment
Bubble wrap from online shopping is light, bulky and rarely recycled. Here it gets a long second life with old cloth that would also be thrown away.

💰 Economy
Two free waste streams make a useful product for camping, school trips or flood relief centres.

🤝 Society
A 2024 student team designed it for people sleeping rough on cold, hard floors: a gift that shows care.

🧺 You need

- Bubble wrap, enough for 18 squares of 30 cm × 30 cm
- Old cloth (bedsheet, curtain or T-shirts) for 18 matching squares
- Ruler, marker, scissors and pins
- Big needle and strong thread, or a sewing machine
- About 10 buttons

🪜 Steps

1. Cut 18 squares of bubble wrap and 18 squares of cloth, each 30 cm × 30 cm.
2. Pin a cloth square onto each bubble-wrap square, bubbles facing in.
3. Sew the pairs into 6 rows of 3 squares.
4. Sew the 6 rows together into one sheet, about 90 cm × 180 cm.
   🔬 The student team made theirs about 100 cm × 180 cm: big enough for an adult to lie on.
5. Fold a strip of cloth over the edges and sew it down so no bubble wrap sticks out.
6. Sew buttons along one long side and the bottom. Cut small button holes (or sew loops) on the matching edges.
7. Test all 3 ways: open flat as a mat (bubble side down), wrap as a blanket, or fold in half and button up as a sleeping bag.
   🔬 The student team's next idea: velcro instead of buttons, so it is quicker to change.

🦺 Safety First

⛔ Plastic over the face can stop breathing. Never let babies or toddlers use or play with it, and never cover the face.

⚠️ Plastic burns and melts easily: keep it away from candles, mosquito coils and stoves.

⚠️ Needles and pins: count them before and after. Keep fingers clear of a sewing-machine needle.

🔬 The science Adults level

This combines two hard-to-recycle streams, LDPE film and mixed textiles, into a long-life product. Plastic does not breathe, so sweat can condense inside: the cloth layer faces the body. Design questions for a real relief product: washability, fire behaviour and how it is collected at end of life.

👩‍🏫 **Teacher tip:** Split the class: cutters, pinners and sewers. Hand-sewing suits ages 10+; a teacher or older pupils use the machine. Two classes can make one bag each for a local shelter or relief centre (ask them first what they need).

🚀 Challenge

1. Warmth test: put a cup of warm water on a square of bubble wrap and another on plain cloth on a cold floor. Measure the temperature every 5 minutes. Which cools faster?
2. Area maths: how many 30 cm squares do you need for a 100 cm × 180 cm bag? (Remember the seams.)

3. Research plastic-bag sleeping mats crocheted from strips of hundreds of bags (the student team's inspiration). Compare time and materials with your bag.

💭 Think about it

- Who in your community could use this, and how would you ask them?
- Why is still air a good insulator?
- What other waste could you use instead of bubble wrap?



The rolled-up bubble-wrap sleeping bag. Photos and posters: UPM chemical engineering students, Faculty of Engineering.

✅ Quick quiz

1. After food, which is the second-largest part of Malaysian household waste?

A. Glass (about 3%)

B. Textiles (about 2%)

C. Plastic (about 22%)

2. A "linear economy" works like this:

A. Make → use → reuse forever

B. Grow → eat → compost

C. Take → make → throw away

3. In a circular economy, materials are...

A. Kept in use for as long as possible

B. Used once, then thrown away

C. Burnt for heat straight away

Answers: 1 C · 2 C · 3 A

Part 8

Hall of Heroes

10 waste-to-wealth projects by UPM chemical engineering students, 2024.



🦸 Zero-Plastic Hero 2024

Zero-Plastic Hero was a service-learning (SULAM) programme of the UPM course ENG3104 Engineers and Society. Chemical engineering students built prototypes from plastic waste and showed secondary school pupils in Selangor how to separate, collect and recycle plastic, and how plastic waste can have real economic value.

400	80	10
pupils and teachers reached	UPM students and lecturers	prototypes

Five of their ideas became WasteQuest labs. Here are all ten heroes' projects.



1. 🧱 Eco-brick Furniture

Bottles packed hard with plastic, glued into a stool with a cushion. The team costed it at about RM 10.50 a stool.



2. 👜 Eco-Viva bag

Plastic bags cut into zig-zag yarn and ironed into a drawstring bag.



3. 🛍️ Tote bags

Tough, water-resistant totes made from foil-lined plastic packaging.



4. 🟫 Coasters & tiles

Plastic waste shaped into hexagon tiles and bottle-cap coasters.

⛔ Showcase only: melting plastic gives off toxic fumes. Try Eco-Bricks (no heat) instead.



5. 🪴 Vertical garden

Hanging bottle planters on a frame, watered by an Arduino moisture sensor and pump.



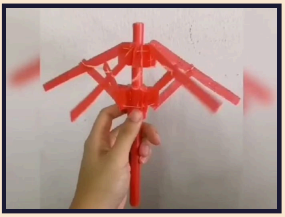
6. 🛌 Sleeping bags

Bubble wrap and cloth squares sewn into a 3-in-1 mat, blanket and sleeping bag for homeless people.



7. 🥬 Hydroponics

Bottle hydroponics inside a mini rain shelter house with a roof of flattened bottles.



8. ☂️ Umbrella & tote bag

Ironed plastic bags as fabric and ironed straws as ribs: a mini umbrella that opens and closes.



9. 🛟 Lifebuoy model

Layers of bubble wrap rolled into a ring and covered with tarpaulin. A model for learning, not a safety device.



10. 🌱 Bio-pots

Plant pots made from home-made bioplastic instead of plastic.

🙏 Advisor and supervisors

Programme advisor: Prof. Ir. Dr. Wan Azlina Wan Ab Karim Ghani

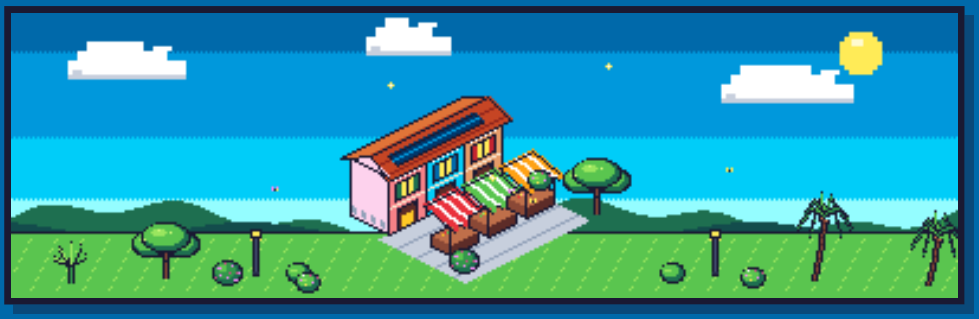
Prof. Madya Dr. Norhafizah Hj. Abdullah · Prof. Madya Dr. Salmiaton Ali · Prof. Madya Datin Ir. Dr. Siti Aslina Hussain · Prof. Madya Dr. Rozita Omar · Prof. Madya Ir. Dr. Shamsul Izhar Siajam · Dr. Nordin Hj. Sabli · Dr. Shafreeza Sobri · Dr. Nur Syakina Jamali · Dr. Mohamad Faiz Mukhtar Gunam Resul · Dr. Halimatun Sakdiah Zainuddin

Thank you to every ENG3104 2024 group and the school that hosted the programme. Photos: the student groups; no photos of school pupils are shown.

Part 6

For Parents, Teachers & Partners

Running WasteQuest with a class, a community or a company.



🤝 Bring WasteQuest to your group

WasteQuest is free. It opens from a link or QR code on any phone, tablet or laptop, with no sign-up, no names and no app to install, and an offline copy works where the signal is weak.

👥 Who it is for

- **Schools and district education offices:** class mode, quizzes, labs and a 20-minute core lesson.
- **Local councils:** Event Town at recycling days and booths, a website embed and a totals-only report.
- **Companies (ESG / CSR):** staff engagement and community events, with reach and learning figures for sustainability reports.
- **NGOs and community groups:** residents' workshops, home missions and the offline copy.
- **Universities:** outreach and service-learning, with a ready before-and-after learning check.
- **Event organisers:** a self-running booth mode with a “next visitor” reset.

🗓️ What a pilot looks like

1. **Demo and scope:** a 30-minute demo; agree the audience, venues, length and what should be reported.
2. **Local check:** the partner checks sorting rules and wording for their area; permissions are agreed (parent consent for under-18s).
3. **Run:** a 4–6 week trial or one school term; players do a 5-question check before and after.
4. **Review:** an evaluation summary, then a joint decision to continue, change or stop.

📊 WasteQuest and ESG reporting

Waste is a standard topic in sustainability reporting (for example GRI 306: Waste 2020), and waste generated in operations is Scope 3, category 5 under the GHG Protocol. Bursa Malaysia requires listed issuers to include a sustainability statement in the annual report. A WasteQuest programme can give coded, name-free figures on reach and learning (before-and-after checks) to support the engagement part of such reports. It does not measure tonnes of waste diverted.

✉️ Contact

Prof. Ir. Dr. Wan Azlina Wan Ab Karim Ghani, Department of Chemical and Environmental Engineering, Faculty of Engineering, Universiti Putra Malaysia. **wanazlina@upm.edu.my**

🎓 Certificates

WasteQuest offers three certificate tracks. Learners complete them on this website at 🏅 **Certificates**. Every certificate is a **Certificate of Completion** with a verification code.

Track	Audience	Assessment	Pass mark	Other requirement
🌱 **Junior Eco-Hero**	Kids 7–12	10 questions from the kids track	**70%**	Earn at least 3 game badges
🏆 **Zero-Waste Champion**	Teens 13–17	15 questions from the teens track	**75%**	At least 5 badges, including at least 1 lab badge
🎓 **Waste-to-Wealth Practitioner**	Adults	20 questions from the adults track	**80%**	At least 2 lab badges + W2W product pitch (portfolio, teacher-verified)

Non-credit Certificate of Completion. No MQA accreditation, quality verification, academic credit or credit transfer is claimed. A formal UPM micro-credential would be a separate route that needs approval through the Faculty and UPM's Centre for Academic Development and Leadership Excellence (CADe-Lead).

Practitioner design: 20 notional learning hours (non-credit)

In Malaysia, 1 credit = 40 notional learning hours (NLH): all the time a typical learner needs, including guided and independent learning and assessment. Twenty hours of learning does not by itself give any credit; credit needs an approved academic route. The Practitioner is designed with reference to the MQA *Guidelines to Good Practices: Micro-credentials* (2020).

Component	Mode	NLH	Evidence
Online learning: Learn chapters 1–7 (adult view)	Independent	4	All 7 quick checks complete (Learn badge)
Games and simulations (Footprint, Waste to Cash, pH, compost, eco-enzyme)	Independent or guided	3	Game badges
Hands-on labs (at least 2, incl. preparation and clean-up)	Face-to-face, supervised	6	Lab badges; product photos and notes
Portfolio and W2W product pitch (research, costing, rehearsal)	Independent and group	5	Completed pitch form
Assessment: final quest (20 questions) and pitch session	Online and face-to-face	2	Quest score; pitch rubric
Total		**20**	**non-credit**

Course learning outcomes (2) and assessment matrix

Two course learning outcomes (CLOs), each covering module LOs as sub-outcomes. This follows good practice of a small number of focused outcomes.

CLO	Sub-outcomes	Activities	Assessment
CLO1 Explain W2W, circular-economy and waste-management principles in the Malaysian context (Act 672, EPR, SDGs).	LO1, LO2, LO3	📘 Learn 1 📘 Learn 2 📘 Learn 3 📘 Learn 4 📘 Learn 5 📘 Learn 6 👣 My Waste Footprint	Final quest, 20 questions (pass 80%)
CLO2 Produce and pitch a safe, costed W2W product that adds environmental, economic and social value.	LO4, LO5, LO6	🔬 Labs 💰 Trash-to-Cash Tycoon 🎤 W2W pitch form	Lab portfolio (2 labs) and W2W pitch rubric

Assessment weighting and completion rule

Assessment	Weight	Minimum to complete
Final quest (20 questions, adults track)	30%	80%
Lab portfolio (2 labs, product rubric)	30%	2 lab badges
W2W product pitch (rubric below)	40%	No criterion at Level 1

Pitch score = total of the six criteria (each 1–4) ÷ 24 × 100. The website checks the quest score, lab badges and a complete pitch form; the facilitator verifies the pitch with this rubric and signs the portfolio sheet.

W2W Product Pitch rubric

Criterion	1 Beginning	2 Developing	3 Proficient	4 Exemplary
Problem and waste stream	Waste stream unclear; no link to a real problem.	Waste stream named; problem described in general terms.	Clear waste stream and local problem, with an estimate of kg diverted.	As Proficient, plus sourced data and a realistic plan to collect the waste.
Product and process	Product unclear or not made.	Product made; steps incomplete.	Working product; steps clear and repeatable; science explained.	As Proficient, plus tested and improved, with evidence (e.g. a second version).
Safety and environmental impact	Hazards not identified.	Some hazards listed; controls incomplete.	Hazards and controls complete; no new pollution created.	As Proficient, plus life-cycle thinking: energy, water and the product's end of life.
Value proposition and costing	No customer or cost.	Customer named; cost or price incomplete.	Unit cost, price and margin calculated; customer need explained.	As Proficient, plus scale-up or "what if costs rise" and comparison with an alternative product.
Social impact	Not considered.	General benefit stated.	Specific community benefit (jobs, awareness, use in schools) linked to an SDG.	As Proficient, plus a plan to involve the community and measure the impact.
Communication	Hard to follow.	Main points clear; over time or few visuals.	Clear, on time, answers questions well.	Persuasive and well structured; uses data and visuals; confident in Q&A.

Quality assurance notes

- MQA defines micro-credentials as certification of assessed knowledge, skills and competencies in a specific field, from accredited programme components or stand-alone courses (MQA, 2020).
- Stand-alone micro-credentials are quality-verified through an **MQA-appointed Quality Verification Centre (QVC)** using form SAMC-01. The minimum is **40 NLH (1 credit)**, in steps of 20 NLH; no more than 2 ILOs per credit; apply within 3 months before the course or cohort ends; verification lasts 3 years (MQA GGP QVSAMC 2023, as summarised in university guidelines).
- **What this means:** at 20 NLH, the Practitioner is planned as a non-credit Certificate of Completion. Who issues it and what approval it needs are still to be confirmed with the Faculty and UPM CADe-Lead. It is below the 40 NLH minimum for MQA quality verification, which is voluntary; any formal micro-credential is a separate, approval-dependent route.
- Before applying, document: market need; LOs aligned to the MQF level; assessment matched to the LOs (at least two methods); facilitator qualifications; and secure records so certificates cannot be forged.
- An official certificate usually also shows a serial number, learner ID, learning outcomes, language, total learning hours, assessment type, grade and an authorised signature. The WasteQuest certificate currently shows name, track, date, score and a verification code.

Next steps (through UPM)

1. Pilot the 20-NLH Practitioner with one adult cohort; collect quest scores, rubrics and feedback.
2. Confirm the issuer and approval for the completion certificate with the Faculty and UPM CADe-Lead.
3. Only if a formal micro-credential is wanted later: develop it through UPM's micro-credential process (CADe-Lead and the offering Faculty). MQA quality verification would need at least 40 NLH.
4. Only after approval, update certificates and this page to say so.

Teacher guide

📋 Overview & outcomes

WasteQuest turns waste education into a quest. Learners find out where Malaysia's waste goes, play games that build sorting and science skills, then make real products from waste in hands-on labs. The module combines gamification (points, badges and certificates), experiential learning, STEM and sustainability education. Its aim is to grow **Zero-Waste Champions**: people who reduce, sort and transform waste at home, at school and at work.

📥 **Downloads:** Booklet (English, PDF) · Booklet (BM, PDF) · 📖 eBook (read online) · eBook (English, PDF) · eBook (BM, PDF) · Offline pack with videos (ZIP, about 215 MB): unzip to a laptop or USB and open WasteQuest_offline.html, no internet needed · 🔗 QR code, website embed and Google Classroom links

What is new in the 2026 edition

- A website that runs on phones, tablets and classroom projectors, in English and Bahasa Melayu.
- Four levels: kids (7–12), teens (13–17), adults, and a teacher view with teaching notes.
- Games, science simulations, 15 hands-on labs with safety controls, a live Class Battle and three certificate tracks.
- Plastic-reuse labs: vertical garden, bottle hydroponics, fused-plastic bags, float ring model and bubble-wrap sleeping mat.
- Science checked and corrected: unsafe or inaccurate methods from earlier student proposals are not used.

Teaching approach

Gamification

Points, badges, levels and certificates keep learners motivated and give instant feedback.

Experiential learning

Learners do, reflect, understand and try again (Kolb's cycle, below).

STEM

Chemistry (pH, fermentation), biology (microbes, compost), maths (data, costing) and engineering design.

Sustainability and ESG

Malaysian waste data, the SDGs, Act 672 and Extended Producer Responsibility (EPR).

Kolb stage	What learners do in WasteQuest
1. Concrete experience	Play a game; make a product in a lab.
2. Reflective observation	Quick checks, class discussion, reflection worksheet.
3. Abstract conceptualisation	Learn chapters and science simulations explain why it works.
4. Active experimentation	Improve the product, write an action plan, pitch a W2W idea.

Objectives

1. Raise awareness of Malaysia's waste challenge and the waste hierarchy through interactive activities.
2. Introduce Waste-to-Wealth and the circular economy through experiential, hands-on learning.
3. Strengthen STEM learning through safe, science-based waste conversion projects.
4. Encourage sustainable lifestyles, environmental stewardship and green entrepreneurship in daily life.

Learning outcomes, mapped to activities and assessment

By the end of the module, learners will be able to... The last column shows which of the four 2025 learning outcomes each one extends.

LO	Outcome	Level	Activities	Assessment	2028 SLO
LO1	Explain sustainability, the waste hierarchy, Waste-to-Wealth (W2W) and the circular economy, using Malaysian waste data.	Understand	📘 Learn 1 📘 Learn 2 📘 Learn 3 📘 Learn 4 🕵️ Myth or Fact? ✏️ Let's Cross Together!	Learn quick checks; quiz	LO1

LO	Outcome	Level	Activities	Assessment	202S LO
LO2	Separate household waste correctly into the SWCorp blue, orange and brown bins and explain why separation at source matters.	Apply	📘 Learn 5 🗑️ Sort It Out! 🃏 Eco-Match 🔎 Fun Puzzle: Word Search	Sort It Out score; quiz	LO4
LO3	Explain the science behind a waste conversion: composting, fermentation, natural pH indicators or bioplastics.	Understand, analyse	🌱 Compost Master 🍊 Eco-Enzyme Mixer 🧪 Say No to Litmus! 🌈 Say No to Litmus: Rainbow pH from Kitchen Waste	Reflection worksheet; quiz	LO1
LO4	Make a W2W product safely by following a method and its safety controls.	Apply (hands-on)	🔬 All 15 labs	Product rubric; lab badge	LO2

LO	Outcome	Level	Activities	Assessment	2028 LO
LO5	Evaluate the environmental, economic and social value of a W2W product, including its cost and price.	Evaluate, create	💰 Trash-to-Cash Tycoon 👣 My Waste Footprint 📘 Learn 4 📘 Learn 6 🎤 W2W pitch form	W2W pitch rubric	LO2
LO6	Reflect on personal habits and promote zero-waste practices at home, in school and in the community.	Value (affective)	📘 Learn 7 🥁 Class Battle	Action plan; reflection rubric	LO3, LO4

Benefits of Waste-to-Wealth

🌿 Environmental	💰 Economic	🤝 Social
Less pollution of air, rivers and sea	Value-added products from low-cost waste	Greater environmental awareness
Lower greenhouse gas emissions, especially methane from landfills	Green entrepreneurship and small businesses	Sustainable lifestyles and habits
Conserves raw materials, water and energy	Lower disposal costs; new jobs	Creativity, innovation and teamwork

🗓️ Session plans

Four ready-to-run plans. Each activity links to the page you need; the right-hand column gives a question to ask. Adjust timings to your group.

⏱️ 60-minute lesson

Kids 9–12 or teens · one class period (split into two if periods are 30–40 min)

Materials: Projector; three boxes coloured blue, orange and brown; 10 clean waste items; natural pH indicator prepared by the teacher; small cups of safe test liquids (vinegar, lime juice, baking-soda water, soapy water); exit tickets.

Time	Activity	Open	Facilitator prompt
0:00–0:05	Hook: "What did you throw away today?" Show the 1.17 kg stack.	📘 Learn 1	*"If each of us throws away about 1.17 kg a day, how much does our class throw away in a year?"*
0:05–0:20	Learn chapters 1 and 5 on the projector; answer the quick checks by show of hands.	📘 Learn 1 📘 Learn 5	*"Which bin is for paper? Why must food waste stay out of the recycling bins?"*
0:20–0:35	Play Sort It Out in pairs, or a 10-question Class Battle.	🗑️ Sort It Out! 🥁 Class Battle	*"Before you drop it, say the bin and the reason."*

Time	Activity	Open	Facilitator prompt
0:35–0:52	Hands-on demo: groups test safe liquids with the natural pH indicator and record the colours, then try the pH lab.	🌈 **Say No to Litmus: Rainbow pH from Kitchen Waste** 🧪 **Say No to Litmus!**	*"Which liquids are acids? How can a plant scrap tell us?"*
0:52–1:00	Exit ticket: one thing I learned and one action I will take.	📘 **Learn 7**	*"What will you refuse or reduce this week?"*

⚠️ **Adult supervision required.** The teacher prepares the indicator with hot water before class. Learners use only the safe liquids listed: never bleach, drain cleaner or oven cleaner. Do not taste anything.

🛠️ Workshop: 2 hours 20 minutes

Teens, university students or community groups · the original 2025 module flow, improved

Materials: Projector and internet (for Class Battle); devices for games (one per pair); lab materials from each lab page; printed product rubrics and reflection worksheets; first-aid kit; gloves, aprons, paper towels.

Time	Activity	Open	Facilitator prompt
0:00–0:15	Briefing: welcome and objectives; groups of 4–5 with roles (leader, materials, safety officer, recorder, presenter); safety briefing with the lab rules checklist.	📎 **Safety**	*"Our safety officer can stop the group at any time if something looks unsafe."*
0:15–0:35	Presentation: waste in Malaysia, the waste hierarchy, circular economy and W2W.	📘 **Learn 1** 📘 **Learn 2** 📘 **Learn 3** 📘 **Learn 4**	*"Where does most household waste in Malaysia end up? How could it become a resource?"*
0:35–1:05	Quiz and games: station rotation (8 min each) of Sort It Out, Memory Match and Myth or Fact, then a 6-minute Class Battle.	🗑️ **Sort It Out!** 🃏 **Eco-Match** 🕵️ **Myth or Fact?** 🏟️ **Class Battle**	*"Which myth surprised you the most?"*

Time	Activity	Open	Facilitator prompt
1:05–2:05	Hands-on: each group completes one lab and scores it with the product rubric. Low-heat choices: ecobricks, natural pH indicator paper, odour absorber. Candle or bioplastic only with an adult at each stove.	🧱 **Eco-Bricks & Eco-Bags** 🌈 **Say No to Litmus: Rainbow pH from Kitchen Waste** ☕ **Kopi Wira Bau: Coffee-Ground Odour Absorbers** 🕯️ **Eco-Candles from Used Cooking Oil** 🌽 **Cornstarch Bioplastic**	*"What waste did you use, what did you make, and what science made it work?"*
2:05–2:20	Reflection and assessment: a 1-minute pitch per group, the reflection worksheet, and the quiz or certificate quest.	⚡ **Eco Quiz Blast** 🏅 **Certificates**	*"Who would use or buy your product, and how much waste could it save?"*

⚠️ **Adult supervision required.** Heat (candle, bioplastic) and sharp tools need an adult at every station. Prepare materials and cut bottles before the session.

🌱 Eco-Club: 4 weeks

Primary or secondary school club · one 60–90 minute meeting a week

Materials: Gloves and a kitchen scale for the bin audit; a plastic bottle for eco-enzyme; a compost bin; lab materials; space for a small exhibition.

Time	Activity	Open	Facilitator prompt
Week 1	Waste detectives: Learn 1–2; Sort It Out; audit one day of the class bin (weigh by type, gloves on); start an eco-enzyme bottle, which needs 3 months.	📘 Learn 1 📘 Learn 2 🗑️ Sort It Out! 🍊 Eco-Enzyme	*"What surprised you in our bin? Which item could have been avoided?"*
Week 2	Science of change: Learn 3–4; compost, eco-enzyme and pH simulations; natural pH indicator lab; set up a compost bin.	📘 Learn 3 📘 Learn 4 🌱 Compost Master 🍊 Eco-Enzyme Mixer 🧪 Say No to Litmus! 🌈 Say No to Litmus: Rainbow pH from Kitchen Waste 🪱 Composting: Bokashi & Worms	*"What do microbes need to break down food waste?"*

Time	Activity	Open	Facilitator prompt
Week 3	Makers: one product lab per group, then cost the product in Waste to Cash.	🎁 **Trash to Treasure: Soap, Tote & Toy Car** 🌱 **Self-Watering Bottle Garden** 🌽 **Cornstarch Bioplastic** 🕯️ **Eco-Candles from Used Cooking Oil** 💰 **Trash-to-Cash Tycoon**	*"What does one item cost to make? Who would buy it, and for how much?"*
Week 4	Champions: Learn 5–7; action plans; Class Battle tournament; mini exhibition for parents or the school; certificate quests.	📘 **Learn 5** 📘 **Learn 6** 📘 **Learn 7** 🏟️ **Class Battle** 🏅 **Certificates**	*"What will our club do next term to keep the change going?"*

⚠️ **Adult supervision required.** Eco-enzyme bottles build up gas: use plastic bottles only, never sealed glass, and release the gas daily in the first month. Cover any standing water so Aedes mosquitoes cannot breed.

💼 Adult / ESG half-day: 3.5 hours

Staff, community leaders, CSR and sustainability teams · counts towards the Practitioner Certificate of Completion (non-credit)

Materials: Projector and internet; one device per pair; lab materials for three labs; printed pitch rubric; aprons and gloves.

Time	Activity	Open	Facilitator prompt
0:00–0:20	Opening: Malaysia's waste picture and why it matters to organisations (set the audience to Adults).	📘 Learn 1	*"What waste does your organisation produce, and who pays for it?"*
0:20–0:50	Circular economy, the waste hierarchy and Extended Producer Responsibility (EPR).	📘 Learn 2 📘 Learn 3 📘 Learn 4	*"Where on the hierarchy are your current practices? What would moving one step up take?"*
0:50–1:20	Simulations in pairs: carbon footprint and Waste to Cash costing.	👣 My Waste Footprint 💰 Trash-to-Cash Tycoon	*"What price makes the product viable without overclaiming its green benefits?"*
1:20–1:35	Break		

Time	Activity	Open	Facilitator prompt
1:35–2:35	Hands-on lab: choose one of three.	🕯️ **Eco-Candles from Used Cooking Oil** 🍊 **Eco-Enzyme** 🌽 **Cornstarch Bioplastic**	*"What would it take to make 100 of these a month, safely?"*
2:35–3:00	Pitch preparation with the W2W pitch form.	🎤 **W2W pitch form**	*"State the waste stream, the customer, the unit cost and one SDG."*
3:00–3:20	Pitches (3 minutes each) with peer scoring on the pitch rubric. For more than 6 groups, run a gallery walk.	📎 **Certificates**	*"What is the strongest point of this pitch, and one question you would ask an investor?"*
3:20–3:30	Close: Practitioner quest (20 questions) now or as follow-up; next steps.	🏅 **Certificates**	*"What one change will your team make next month?"*

⚠️ **Adult supervision required.** Hot wax and hot starch mixtures cause burns: one facilitator per stove, long sleeves and closed shoes.

Safety

⚠️ **Adult supervision required.** Heat, sharp tools, fermentation gas and electricity are used in some labs. Read each lab page before the session and prepare materials in advance.

Lab rules checklist

- ☐ An adult supervises every lab. For heat or sharp-tool steps, one adult per group.
- ☐ Safety briefing before starting; each group names a safety officer.
- ☐ Tie back long hair; closed shoes; apron. Goggles for hot liquids and pH tests.
- ☐ Use only clean waste. No broken glass, needles, batteries, medicines or chemical containers.
- ☐ Gloves for food scraps, fish waste and compost. Wash hands with soap afterwards.
- ☐ Never taste or eat anything in a lab, including the products.
- ☐ Adults handle stoves, hot wax, hot water and irons. Never leave heat unattended. Never put water on a wax or oil fire: turn off the heat and cover it with a lid.

- ☐ Adults pre-cut bottles with craft knives; children use safety scissors.
- ☐ Never melt or burn plastic in a pot, over a flame or in oil. The only plastic heating allowed: an adult ironing PE bags between baking paper in a ventilated room (Fused-Plastic lab).
- ☐ No lye (sodium hydroxide) with children. Adult groups making soap follow chemical-safety rules.
- ☐ Ask about allergies (fish, coffee, fragrance, latex gloves) before the lab.
- ☐ Label every jar or bottle with contents, name and date.
- ☐ Electronics: low-voltage USB or battery only, kept away from water.
- ☐ First-aid kit ready; emergency number 999. Clean up and sort leftover waste into the right bins.

🚫 **Never use:** the 2024 student method of melting plastic in hot engine oil. It gives off toxic fumes and is a serious fire risk.

Risk table for the 18 labs

Risk ratings are our guide for a typical school session with the controls in place. Review them for your own space and group.

Lab	Hazard	Risk	Control	Supervision
🕯️ Eco-Candles from Used Cooking Oil	Hot wax or oil; flame	High	Adult melts wax in a double boiler, never over direct flame; learners add colour and fragrance away from the stove; no water on a wax fire (lid on); light candles only when set and supervised.	Adult does all heating
🐟 Fish-Waste Pet Treats	Raw fish (bacteria); knives; stove heat	Medium	Gloves; separate board; adult cuts and cooks thoroughly; wash hands and surfaces; keep chilled; check fish allergies.	Adult cuts and cooks
🎁 Trash to Treasure: Soap, Tote & Toy Car	Scissors and needles; hot iron and fumes if fusing plastic bags	Medium	Safety scissors; needle count before and after; adult irons between baking paper on a low setting in a ventilated room; stop if it smokes.	Adult irons
🌈 Say No to Litmus: Rainbow pH from Kitchen Waste	Hot water for extraction; test liquids	Low	Adult pours hot water; use only food-safe liquids (vinegar, lime juice, baking-soda water, soapy water); never bleach, drain or oven cleaner; goggles; do not taste.	Adult pours hot water
☕ Kopi Wira Bau: Coffee-Ground Odour Absorbers	Mould on damp grounds; oven heat	Low	Dry grounds fully (sun or adult-operated oven); throw away mouldy batches; use breathable bags.	Adult uses the oven

Lab	Hazard	Risk	Control	Supervision
🌱 **Self-Watering Bottle Garden**	Cut bottle edges; standing water (Aedes mosquitoes); electricity in the sensor version	Medium	Adult cuts with a craft knife and tapes edges; cover the reservoir and check it weekly; low-voltage electronics only, kept dry.	Adult cuts; weekly check
🍊 **Eco-Enzyme**	Gas pressure from fermentation; not drinkable; ants	Medium	Plastic bottles only, never sealed glass; leave headspace; release gas daily in the first month; label “Not for drinking”; store out of reach; dilute before use and patch-test on skin.	Adult checks weekly for 3 months
🧱 **Eco-Bricks & Eco-Bags**	Sharp plastic pieces; packing stick	Low	Only clean, dry soft plastic (dirty plastic grows mould); use a blunt wooden stick; gloves optional.	Normal supervision
🪱 **Composting: Bokashi & Worms**	Microbes and fungal spores; pests; garden tools	Medium	Gloves and handwashing; no meat, dairy or oily food (attracts pests); keep the bin covered; learners with asthma or weak immunity do not turn dry compost.	Adult present when turning
🌽 **Cornstarch Bioplastic**	Stove heat; hot sticky mixture can spatter	Medium	Adult heats on low and stirs with a long spoon; goggles; cool fully before shaping.	Adult does all heating

Lab	Hazard	Risk	Control	Supervision
🪴 **Bottle Vertical Garden**	Cut bottle edges; heavy wet soil overhead; standing water (Aedes); money plant is toxic if eaten	Low	Adult cuts and tapes edges; hang on a strong frame away from walkways; empty trays and check weekly; wash hands; edible plants and money plant kept apart and labelled.	Adult cuts
🥬 **Bottle Hydroponics**	Nutrient salts (not for drinking); standing water (Aedes); cut edges, nails and hammer	Medium	Adult measures nutrients from the label; label bottles “Not for drinking”; seal gaps with tape and check for wrigglers weekly; adult cuts and hammers; spent solution poured on plants, not drains.	Adult handles nutrients and tools
☂️ **Fused-Plastic Bags & Mini Umbrella**	Hot iron; plastic fumes; needles	High	Adult irons only, low-medium with no steam, baking paper on both sides, windows open; PE bags (2 or 4) only, never PVC (3) or foil-lined wrappers; stop if it smokes or smells; needle count.	Adult does all ironing
🛟 **Bubble-Wrap Float Ring (Model)**	Drowning risk at the water basin; model mistaken for a real float; plastic film (suffocation for toddlers)	Medium	Adult stays at the basin and empties it after; say clearly it is a model, never for swimming or floods; teach “Reach or throw, don't go” and call 999; keep plastic away from toddlers.	Adult at the water

Lab	Hazard	Risk	Control	Supervision
🛌 **3-in-1 Bubble-Wrap Sleeping Bag**	Needles and sewing machine; plastic over the face (suffocation); plastic catches fire	Low	Needle count; adult or older pupils use the machine; label the finished bag: not for babies or toddlers, never over the face, keep away from flames and mosquito coils.	Normal supervision; adult on the machine

📝 Rubrics & reflection

For kids and teens doing a hands-on lab. Score each criterion 1–4 (total out of 12). Teens can also peer-assess another group.

Hands-on rubric

Criterion	1 Beginning	2 Developing	3 Proficient	4 Exemplary
Product	Not finished.	Finished with a lot of help; some steps followed.	Finished; all steps followed; works as planned.	Works well and shows a creative improvement.
Teamwork and safety	Works alone or off task.	Joins in when asked.	Does own role, shares, follows safety rules.	Helps others, includes everyone, leads safely.
Reflection	Says only what was made.	Says what was made and from which waste.	Explains the science or why it helps the environment.	Also suggests an improvement and a real-life use.

Kids' star self-check

I can…	Colour the stars
I made my product safely.	☆ ☆ ☆
I followed the steps.	☆ ☆ ☆
I helped my team.	☆ ☆ ☆
I can say what waste we used.	☆ ☆ ☆
I can say how it helps the Earth.	☆ ☆ ☆

📝 Reflection worksheet

Name / group / date

What waste did we use, and where would it normally go?

What did we make? Draw or describe it.

What science made it work? (e.g. microbes, acids and bases, heat)

What went well? What would we change next time?

How much waste could this save in a month? Show your working.

Who could use or buy it?

My zero-waste promise for this week:

🏫 Curriculum links

WasteQuest supports, but does not replace, the national curriculum. The links below are general; always check the current DSKP for your year and subject, because standards and chapter numbers are revised.

Level	Subject and topic	WasteQuest activities
Primary (KSSR)	Science, upper primary: micro-organisms, and preservation and conservation (Pemeliharaan dan Pemuliharaan) in Year 6. Check your DSKP.	🌱 Compost Master 🪱 Composting: Bokashi & Worms 🗑️ Sort It Out!
Form 1 (KSSM)	Design and Technology (RBT): Ch 3 Design Process, Ch 4 Sketching, Ch 5 Technology Application.	🌱 Self-Watering Bottle Garden 🎁 Trash to Treasure: Soap, Tote & Toy Car 🎤 W2W pitch form
Form 2 (KSSM)	Science: Ch 2 Ecosystems; Ch 6 Acids and Alkalis.	🌱 Compost Master 🧪 Say No to Litmus! 🌈 Say No to Litmus: Rainbow pH from Kitchen Waste 🍊 Eco-Enzyme
Secondary (KSSM)	Geography: environment and sustainability topics (check the DSKP for your form).	📘 Learn 1 📘 Learn 6 👣 My Waste Footprint

Level	Subject and topic	WasteQuest activities
All KSSM subjects	Cross-curricular elements (EMK): Environmental Sustainability, Entrepreneurship, Creativity and Innovation, Science and Technology.	💰 Trash-to-Cash Tycoon 🏟️ Class Battle
University and adults	Sustainability, engineering and society, ESG reporting (e.g. GRI 306: Waste), EPR, SDG 12.	📘 Learn 3 📘 Learn 4 📘 Learn 6 🎤 W2W pitch form

These are suggested links only. Please check them against the DSKP edition your school uses.

Running Class Battle

Class Battle is a live quiz for the whole class: the questions show on the projector and students answer on their phones. **Class Battle**

Steps

1. Open WasteQuest on the computer connected to the projector and go to Class Battle.
2. Choose **Host a class**. Pick the level (kids, teens, adults), topic, number of questions and seconds per question.
3. Students scan the QR code, or open WasteQuest → Class → Join a class and type the 5-character code. No app or login; nicknames only.
4. When everyone is in, press **Start quiz**. Faster correct answers earn more points; the leaderboard shows after each question.
5. At the end, download the results (CSV). The "% correct" per question shows which topics to teach again.

Needs internet. Live mode links phones to the projector through a free online service (PeerJS). It does not work when WasteQuest is opened from a file or localhost: use the online site. School Wi-Fi may block it; a phone hotspot usually works.

No phones or no internet? Use **Team mode**: 2–4 teams play on the projector only, offline. After each question, tap every team that got it right (+100 points).

Teaching tips

- Warm-up: 5 questions at the start to find out what the class already knows.

- Exit ticket: 5 questions at the end; reteach the question with the lowest % correct next lesson.
- Remind students to use kind nicknames; remove any unsuitable name from the lobby.
- Pause after each question to explain the answer: the discussion is where most learning happens.

🤝 Who can use it

Who can use WasteQuest

User	Typical use	Suggested plan
Primary and secondary schools	Science, RBT and Eco-Club lessons; school recycling drives	60-minute lesson; Eco-Club
Universities and TVET	Sustainability and engineering-and-society courses; student outreach projects	Workshop; Practitioner
Government agencies and local authorities	Public awareness on separation at source and recycling	Workshop; community package
NGOs and community learning centres	Community workshops and W2W enterprise ideas	Workshop; Eco-Club
Training providers	Sustainability and ESG courses	Adult half-day; Practitioner
Companies and green-technology organisations	Staff ESG training; CSR programmes with schools	Adult half-day; Practitioner

Packages

Package	What it includes
📱 Interactive digital learning module	Free bilingual website for self-directed and classroom learning: Learn chapters, games, simulations, labs, Class Battle, certificates.
🏫 School environmental education toolkit	60-minute lesson, Eco-Club plan, rubrics, reflection worksheet, safety checklist.
🎒 Sustainability educational package	Workshop with facilitator, lab materials and certificates.
💼 ESG training module	Adult half-day and the 20-hour Practitioner Certificate of Completion (non-credit).

Package	What it includes
Community sustainability learning package	Workshop plus community labs (compost, eco-enzyme, ecobricks).

The website is free. Facilitated workshops, kits and adult courses are quoted separately, based on group size, activities, materials, venue and travel; duration, certificate and fee are confirmed in the quotation. Bookings are not open yet.

🙏 Credits & citation

Authors

Wan Azlina Wan Ab Karim Ghani, Shafreeza Sobri, Izzudin Ismail, Nur Syakina Jamali, Mohd Faiz Gunam Rasul, Salmiaton Ali
Faculty of Engineering, Universiti Putra Malaysia

Acknowledgements

Many activities grew from projects by UPM engineering students. Their ideas were checked, made safe and adapted for this module.

How to cite

Wan Ab Karim Ghani, W. A., Sobri, S., Ismail, I., Jamali, N. S., Gunam Rasul, M. F., & Ali, S. (2026). *WasteQuest: Waste-to-Wealth (W2W) educational module* [Interactive web module]. Universiti Putra Malaysia, Faculty of Engineering. https://wastequest.github.io/

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💡 Did you know?

Red wigglers live near the surface and eat large amounts of organic waste; their castings are a rich fertiliser.

Part 7

Words, Index & Credits



🔤 Glossary

Each word with its Bahasa Melayu name.

Act 672 *(Akta 672)*: The Solid Waste and Public Cleansing Management Act 2007, which includes separating waste at source.

Aerobic *(Aerobik)*: With oxygen.

Anaerobic *(Anaerobik)*: Without oxygen.

Bioplastic *(Bioplastik)*: Plastic made from plants, for example starch, instead of petroleum.

Buoyancy *(Keapungan)*: The upward push of water that makes things float.

Circular economy *(Ekonomi kitaran)*: An economy that keeps materials in use and designs out waste and pollution.

Compost *(Kompos)*: A dark, crumbly soil improver made when microbes break down food and garden waste with air.

Eco-enzyme *(Eko-enzim)*: A liquid made by fermenting sugar, fruit peels and water (1:3:10) for about 3 months.

Ecobrick *(Eko-bata)*: A plastic bottle packed hard with clean, dry plastic, used as a building block.

ESG *(ESG)*: Environmental, social and governance: how an organisation manages these risks and impacts.

Fermentation *(Penapaian)*: Microbes breaking down sugar without oxygen, making acids, gas or alcohol.

Greenhouse gas *(Gas rumah hijau)*: A gas that traps heat in the atmosphere, such as carbon dioxide and methane.

Hydroponics *(Hidroponik)*: Growing plants in water with nutrients, without soil.

Indicator *(Penunjuk)*: A substance that changes colour with pH, like purple-cabbage or bunga telang juice.

Insulation *(Penebat)*: Material that slows the movement of heat, often by trapping still air.

Landfill *(Tapak pelupusan)*: A site where waste is buried in the ground.

Leachate *(Larut resap)*: Polluted liquid that drains out of waste in a landfill.

Methane *(Metana)*: A greenhouse gas (CH_4) made when food waste rots without oxygen; about 28 times stronger than CO_2 over 100 years.

Micro-credential *(Mikro-kredensial)*: A short, focused course with assessed learning outcomes and a certificate.

Microplastics *(Mikroplastik)*: Plastic pieces smaller than 5 mm.

Open burning *(Pembakaran terbuka)*: Burning waste in the open air; an offence in Malaysia (Environmental Quality Act 1974, s.29A).

pH *(pH)*: A scale from 0 to 14 that shows how acidic or alkaline a liquid is; 7 is neutral.

Recycling *(Kitar semula)*: Processing used materials into new raw materials.

Saponification *(Saponifikasi)*: The reaction of fat or oil with a strong alkali that makes soap.

Scope 3 emissions *(Pelepasan Skop 3)*: Indirect emissions in an organisation's value chain; waste from operations is Scope 3, category 5.

SDGs *(SDG)*: The UN Sustainable Development Goals: 17 global goals for 2030.

SWCorp *(SWCorp)*: The Solid Waste Management and Public Cleansing Corporation, the Malaysian agency for solid waste.

Upcycling *(Kitar naik)*: Turning waste into a product of higher value.

Used cooking oil (UCO) *(Minyak masak terpakai)*: Oil left after frying; it can become candles, soap or biodiesel.

Vermicompost *(Vermikompos)*: Compost made with the help of worms.

Waste hierarchy *(Hierarki sisa)*: Waste options ranked from best (refuse, reduce) to last choice (disposal).

Waste-to-wealth (W2W) *(Sisa kepada kekayaan (W2W))*: Turning waste into products or resources with value.

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🙏 Credits & sources

Written by: Wan Azlina Wan Ab Karim Ghani, Shafreeza Sobri, Izzudin Ibrahim, Nur Syakina Jamali, Mohamad Faiz Mukhtar Gunam Resul, Salmiaton Ali. Department of Chemical and Environmental Engineering, Faculty of Engineering, Universiti Putra Malaysia.

Lab ideas, photos and posters: UPM chemical engineering students (ENG3104 Engineers and Society, 2024 and 2025). Lab guides checked and corrected by the WasteQuest team.

Videos: made by the WasteQuest team in 2026 from the students' demonstrations, narrated with synthetic voices. **Pixel art:** original drawings made for WasteQuest.

📚 Main sources

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- Global Ecobrick Alliance; US CDC (Aedes life cycle); WSAVA (pet treats); IMO LSA Code (lifebuoys); Kratky (2009) hydroponics.
- Every lab and lesson lists its own sources on the website.

🧩 Software and Fonts

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WasteQuest

A free bilingual game for learning to sort, reduce and reuse waste.



Play WasteQuest
wastequest.github.io

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